Supplementary Information: Non-Equilibrium Probes Reveal the Limits of Specialist and Foundation Machine-Learned Interatomic Potentials

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Supplementary Information Roadmap

This Supplementary Information is organized to make the benchmark traceable from reference calculations to model evaluation. Section 1 describes the first-principles setup, NEB reference protocol, MACE training configurations, dataset provenance, and split controls. Section 2 provides the supporting analyses: RMSE benchmarking, structural and thermodynamic validation, molecular-dynamics property calculations, representation-learning descriptors, migration-path diagnostics, self-consistent MLIP-NEB tests, and SHAP surrogate interpretation. Together, these sections separate three observables that are otherwise easy to conflate: the DFT reference barrier on a converged path, the MLIP energy assigned to a fixed DFT path, and the barrier obtained after a model relaxes its own path.

Audit trail.

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1 Detailed Computational Methods

1.1 First-Principles Reference Calculations

All first-principles calculations were performed using the QUANTUM ESPRESSO package [1]. The exchange–correlation energy was described using the Perdew–Burke–Ernzerhof (PBE) functional within the generalized gradient approximation (GGA) [2]. Long-range dispersion interactions, essential for accurately representing the interlayer physics of Sb_2Te_3, were included through the Grimme DFT-D3 correction scheme. Spin–orbit coupling (SOC) was incorporated through fully relativistic ONCV pseudopotentials (SG15 v1.2).

A plane-wave kinetic energy cutoff of 100 Ry and a charge density cutoff of 400 Ry were used for all calculations. Electronic occupations were treated with the cold-smearing scheme using a smearing width of 0.0147 Ry, which provides stable convergence for narrow-gap chalcogenides. Electronic states were iterated until the total electronic energy converged below 10^{-6} Ry, using a mixing parameter of 0.4 and a maximum of 200 electronic iterations per self-consistent cycle.

All structural relaxations were performed using the Broyden–Fletcher–Goldfarb–Shanno (BFGS) algorithm for both ionic and cell degrees of freedom, with force and pressure convergence thresholds of 10^{-3} Ry/Bohr and 0.5 kbar, respectively. A uniform Monkhorst–Pack mesh of $4 \times 4 \times 1$ was employed to sample the Brillouin zone for relaxations and Nudged Elastic Band (NEB) calculations. The $4 \times 2 \times 1$ supercell with dopant contained 81 atoms, corresponding to a two-quintuple-layer (2QL) slab of Sb_2Te_3 with one interstitial Cr atom, separated by sufficient vacuum to avoid spurious periodic interaction. Restart files were written periodically to ensure computational stability during long relaxations.

The training dataset was generated from ab initio molecular dynamics (AIMD) simulations performed in the NVT ensemble using a Langevin thermostat. These simulations covered a range of temperatures (300 K, 600 K, 1200 K) and Cr doping concentrations to ensure the model was exposed to a diverse set of thermodynamic and structural configurations. Each AIMD trajectory was run for 10 ps on a system of 120 atoms. To investigate the nature of atomic migration pathways, minimum energy paths (MEPs) and energy barriers were calculated using the Nudged Elastic Band (NEB) method. In the accompanying data release, each NEB path is reported with its task label, number of images, input structure provenance, restart lineage, convergence threshold, achieved maximum force, and activation barrier. Paths that are not yet formally converged are marked as candidate references and are excluded from final ranking until the corresponding restarted QE calculation reaches the stated criterion. The path-family classification and the associated DFT-quality audit are preserved in the SI audit folder. NEB paths are evaluation/reference data and were not merged into the FT-600K or FT-Multi$_T$ training splits. The released NEB-to-training exporters now fail closed when real QE forces are unavailable, so a NEB geometry cannot silently acquire fabricated force labels.

1.2 MACE Model Training Protocols

1.2.1 Training Hyperparameters

All training and fine-tuning procedures were performed with consistent hyperparameters to ensure direct comparability. Unless otherwise stated, we used the Adam optimizer with an initial learning rate of 1×10^{-3}, a batch size of 32, a maximum of 1000 epochs, and early stopping based on validation-set force MAE. A 10% validation split was applied to all datasets.

Fine-tuning was initialized from the `MACE-MP-0 (PBE)` [3] foundation model. All model parameters were updated without freezing any layers, and all output heads were fine-tuned using the standard multi-head strategy. Stochastic Weight Averaging (SWA) was activated from epoch 100, and Exponential Moving Average (EMA) tracking ($\alpha = 0.99$) was applied throughout training. Energy and forces were read using the keys `energy` and `forces`, respectively, with force, energy, and stress loss weights of 100.0, 1.0, and 0.0. The training configuration employed $l_{\max} = 2$, cutoff radius $r_{\max} = 6.0$ Å. Every training dataset is registered in the MigrationBench manifest with its source trajectory, frame-selection rule, split assignment, random seed, energy/force key mapping, and checksum. The audit covers 9,163 frames: the only nine exact zero-force frames are explicitly labelled one-atom Cr/Sb/Te isolated-atom E_0 references; zero unclassified and zero NEB-derived zero-force frames were found. The complete CSV/SQLite manifest and JSON summary are released with commit `f5c6d19`. This provenance layer makes the energy and force extraction path auditable from raw DFT output to model-training file.

1.2.2 Fine-Tuning Details

The multi-temperature fine-tuning model (FT-Multi_T) was constructed by combining configurations at 300 K, 600 K, and 1200 K. Each configuration was simulated at all three temperatures, ensuring that the Multi_T dataset is strictly balanced across temperatures without statistical or configurational bias. This design probes whether broader thermodynamic sampling improves model robustness and generalization.

The historical frame-level FT-600K model used approximately 5% of the 600 K AIMD pool (about 1000 configurations) and the original frame-level split. For the leakage-controlled A2 re-evaluation, the OMAT-FT-start FT-600K family was retrained on the full trajectory-grouped training partition (1,786 configurations; validation/test partitions of 311/267), while Scratch-5% is a separate 89-configuration low-data control drawn from the same grouped source pool. We keep these protocols distinct and do not use the historical 5% frame-level description to label the grouped FT-600K result. Because preserving complete source trajectories is incompatible with selecting a literal 1000-frame subset without fragmenting the grouped split, the current Scratch-5% result is reported as a grouped low-data control rather than as an exactly matched 1000-configuration ablation. The 600 K data remain representative of realistic thermoelectric operating conditions, lying near two-thirds of the melting point of Sb_2Te_3 and exhibiting significant anharmonic dynamics without structural degradation.

FT-600K, Scratch, and Scratch-5% were retrained under a grouped split in which entire source trajectories are assigned to train, validation, or test, preventing near-duplicate structures from appearing in both training and test partitions (1,786/311/267 frames for the full grouped split; 89/311/267 for Scratch-5%). We additionally audit benchmark-path proximity using nearest-neighbor SOAP cosine distance and Cartesian RMSD between each in-gap NEB image and the training set. For the full grouped training set, the minimum SOAP cosine distance is 5.4×10^{-4} and the minimum comparable RMSD is 2.06 Å; for the Scratch-5% subset, the corresponding values are 6.4×10^{-4} and 2.06 Å. No exact or near-duplicate NEB-image/training-frame matches were found under the declared thresholds (SOAP $\leq 10^{-4}$, RMSD ≤ 0.05 Å) for either split. FT-Multi_T was not included in this grouped-retraining round, so its proximity statistics remain historical (frame-level split) and are not reported here.

2 Supplementary Materials

2.1 MACE Model Performance Evaluation by RMSE

All developed machine-learned interatomic potentials (MLIPs) show low near-equilibrium energy and force errors in the current parity-plot audit (Figure S15), which compares model predictions to the reference DFT calculations. The performance of each model across the training, validation, and test sets is quantitatively summarized in Table S1. These values are used as a baseline pointwise-accuracy diagnostic rather than as final evidence of out-of-trajectory generalization. Table S2 reports the corresponding grouped-split, multi-seed retraining RMSE values and the controlled within-gap barrier-error statistics used to address the overlap, uncertainty, and low-data-control questions.

Table S1: Performance metrics for MACE models across training, validation, and test sets. The FT-600K metrics are the mean ± standard deviation across three independent training runs with different random seeds.

Model	**Train RMSE F** (meV/Å)	**Valid RMSE F** (meV/Å)	**Test RMSE F** (meV/Å)	**Test RMSE E** (meV/atom)
From Scratch	67.1	76.1	75.2	1.0
FT-600K	20.7 ± 11.8	44.6 ± 2.9	37.2 ± 0.6	0.5 ± 0.0
FT-MultiT	20.3	49.1	45.5	0.5

Barrier statistics use the same five DFT images for the controlled within-gap probe as the historical evaluation. FT-Multi_T was not retrained under the grouped protocol in this round and is therefore marked "not retrained" rather than assigned a zero. The path-family table below retains the internal audit IDs, maps them to within-gap, gap-to-QL/into-QL, mixed, or geometry-recheck classes, and records their DFT-quality status; higher-distortion cases are not pooled into a single barrier statistic.

Table S2: **Grouped-split comparison.** Values are means ± sample standard deviations across three independent training initializations unless noted. The within-gap column is the absolute fixed-path error relative to the provisional 0.34 eV DFT candidate. Higher-distortion cases are not ranked; grouped FT-Multi$_T$ was not retrained.

Model	Test E RMSE (meV/atom)	Test F RMSE (meV/Å)	Within-gap error (eV)	Higher-distortion status
FT-600K (OMAT-FT start)	40.7 ± 5.0	259.7 ± 5.8	0.47 ± 0.12	not ranked
Scratch	30.5 ± 3.6	343.6 ± 35.9	1.20 ± 0.27	not ranked
Scratch-5%	16.7 ± 9.2	341.5 ± 15.0	0.68 ± 0.20	not ranked
FT-600K (MPA-0 start; single initialization)	38.7	278.5	0.24	not ranked
FT-Multi$_T$ (grouped)	not retrained	not retrained	not retrained	not ranked

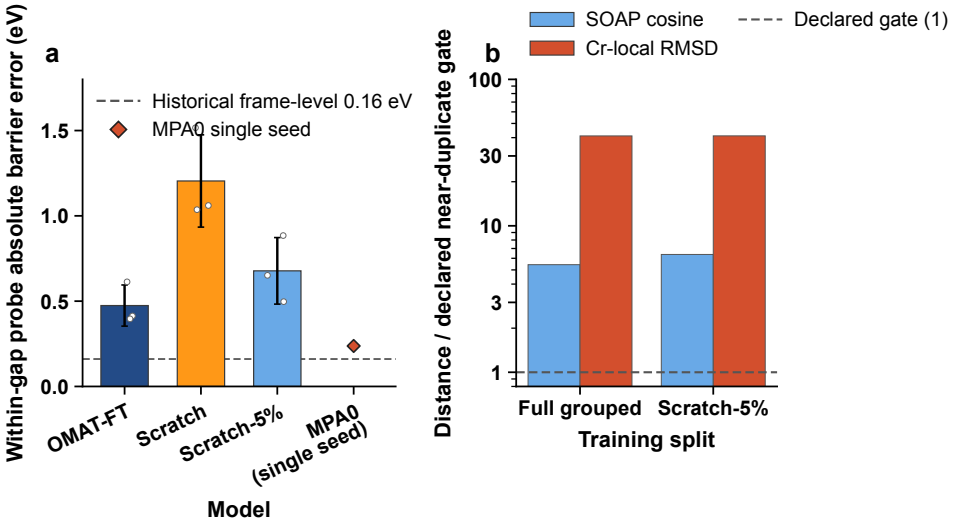


Fig. S1: **Grouped retraining and train–NEB overlap audit.** (a) Absolute fixed-path barrier errors for the controlled within-gap probe under the trajectory-grouped protocol. Open points are the three independent training initializations, bars/whiskers are the mean ± sample standard deviation, and the single-checkpoint MPA0 comparator is shown as a diamond. The dashed line marks the historical frame-level 0.16 eV point estimate, which is not reproduced by the grouped FT-600K evaluation. (b) Minimum training-set SOAP and comparable Cr-local RMSD distances normalized by the declared near-duplicate gates. The dashed line at one is the declared near-duplicate gate; values above one lie outside that gate. RMSD is computed only for species-composition-compatible local environments, while SOAP covers all audited target images.

2.1.1 Task-conditioned distribution diagnostics

The grouped reference distribution was recomputed with a SOAP/PCA fit learned from the grouped training partition only and then applied without refitting to the validation, test, and fixed-path images. The grouped split contains 1,786 training, 311 validation, and 267 test configurations; the training partition contains 1,783 periodic AIMD frames and three isolated-atom calibration frames, while complete source trajectories remain assigned to one partition. The fixed-path images are displayed as a separate geometry class because the audited within-gap images contain 61 atoms, whereas the grouped AIMD cells contain 81–86 atoms. The resulting marginal and smooth two-dimensional densities are shown in the independently rendered reference and MD panels in Fig. S2 and Fig. S4; the original Rockfish validation histories are provided separately in Fig. S3.

Table S3: **Stored MLIP-generated MD configuration-space audit.** The table reports every stored row available to the SOAP/PCA diagnostic. "Short-contact skip" denotes a row retained in the raw inventory but excluded from SOAP because the minimum interatomic distance was below 1.2 Å; it is a geometry-safety status, not a calibrated uncertainty estimate. The common MD velocity seed means that model/checkpoint branches are not independent velocity replicas.

Family	Stored rows	SOAP-valid	Short-contact skip	PCA points
Foundation	4,961	3	4,958	3
Scratch	4,001	1,310	2,691	1,310
FT-600K	4,601	4,601	0	4,601
FT-Multi$_T$	4,001	4,001	0	4,001

The atlas also reports the available grouped-split static test RMSE evidence in the uniform manuscript order Foundation, Scratch, FT-600K, and FT-Multi$_T$. Foundation points are representative zero-shot grouped-test checkpoints, Scratch and FT-600K points are grouped multi-seed observations, and the FT-Multi$_T$ point remains historical frame-level descriptive evidence because grouped FT-Multi$_T$ retraining was not performed in this round. The MD clouds are therefore configuration-space diagnostics rather than DFT errors, and they do not define a universal ranking. In particular, the Foundation and Scratch short-contact statuses are retained rather than being converted into a smooth distribution or interpreted as uncertainty.

The separate failure/CAR figure is an MLIP-only stage-resolved diagnostic based on the 639-candidate failure trajectories. It reports maximum force, paired model disagreement, stage-wise force envelopes, and distance to the grouped training representation. These panels are descriptive diagnostics of failure progression and model disagreement; they are not DFT errors, calibrated uncertainties, or a monotonic scalar OOD ranking. Full source paths, hashes, row counts, and claim boundaries are recorded in the accompanying distribution-atlas manifest.

2.1.2 Seed-aware kinetic uncertainty and protocol audit

The grouped retraining directly addresses initialization uncertainty in the kinetic observable. For the controlled within-gap probe, the absolute fixed-path barrier errors are 0.47 ± 0.12 eV for FT-600K, 1.20 ± 0.27 eV for Scratch, and 0.68 ± 0.20 eV for Scratch-5%, across three independent training initializations. These are sample SDs across training runs rather than calibrated confidence intervals or independent molecular-dynamics replicas. The historical 0.16 eV frame-level point estimate is not reproduced. The path-resolved candidate audit gives Scratch errors of 0.51 ± 0.11 eV and 0.72 ± 0.55 eV on two gap-to-QL/into-QL candidates and 1.77 ± 1.00 eV on a geometry-recheck candidate; the first is comparatively stable within the tested initializations, while the latter two are more seed-sensitive. The corresponding FT-600K values are 9.60 ± 0.57, 10.40 ± 0.44, and 9.70 ± 0.20 eV, respectively. These statistics replace the former untestable "by chance" wording with a class-resolved statement, but remain conditional on provisional DFT references.

Table S4: **Class-resolved kinetic errors.** Entries are absolute fixed-path barrier errors relative to the current pathway-specific DFT candidate. The table summarizes three independent training initializations; run-level observations and audit IDs remain in the accompanying CSV and SI audit table. Higher-distortion references are provisional and one candidate requires a geometry recheck.

Pathway class	Model	Mean ± SD (eV)
Gap-to-QL/into-QL candidate A	Scratch	0.51 ± 0.11
Gap-to-QL/into-QL candidate A	FT-600K	9.60 ± 0.57
Gap-to-QL/into-QL candidate B	Scratch	0.72 ± 0.55
Gap-to-QL/into-QL candidate B	FT-600K	10.40 ± 0.44
Controlled within-gap probe	Scratch	1.20 ± 0.27
Controlled within-gap probe	FT-600K	0.47 ± 0.12
Controlled within-gap probe	Scratch-5%	0.68 ± 0.20
Geometry-recheck candidate	Scratch	1.77 ± 1.00
Geometry-recheck candidate	FT-600K	9.70 ± 0.20
Additional within-gap candidate	Scratch	0.33 ± 0.06
Additional within-gap candidate	FT-600K	0.21 ± 0.18

The completed broader protocol matrix is retained as a separate diagnostic rather than pooled with the strict grouped lineage. The fixed-path matrix contains 50 finite cells for the 61-atom system and 50 for the 81-atom system, with 30 force-pathology flags across the 100 cells. The 81-atom chain uses high-force candidate references. The self-relaxed matrix contains 18 optimizer-converged cells, 17 finite but non-converged cells, and 5 legacy diagnostics with incomplete convergence fields. Self-relaxed values are MLIP-native barrier proxies and the relaxed images have not been DFT-rescored; these rows therefore document protocol stability and failure modes rather than physical barrier accuracy.

Table S5: **Supporting 61-atom fixed-path protocol matrix.** Values are absolute errors from the current matrix and are shown as mean ± sample SD across seeds 123/234/345. Scratch-Multi$_T$ is a separate lineage from the strict grouped Scratch arm. Force-pathology cells are retained and are not silently treated as accurate barriers.

Family	Audit ID (SI only)	Mean (eV)	SD (eV)	Runs	Force-pathology cells
FT-600K	1-2	8.81	0.43	3	3
FT-600K	1-3	10.73	1.04	3	3
FT-600K	1-4	0.86	0.07	3	0
FT-600K	1-5	10.77	1.04	3	3
FT-600K	1-7	0.42	0.07	3	0
Scratch-Multi$_T$	1-2	13.50	21.84	3	2
Scratch-Multi$_T$	1-3	27.68	35.93	3	3
Scratch-Multi$_T$	1-4	0.50	0.17	3	0
Scratch-Multi$_T$	1-5	12.11	11.79	3	3
Scratch-Multi$_T$	1-7	0.21	0.15	3	0

A deeper analysis of the training dynamics reveals important characteristics of the models. The model trained from scratch exhibits a relatively small gap between training (67.1 meV/Å) and validation (76.1 meV/Å) force RMSEs. While this suggests only moderate overfitting, its high error across all datasets indicates that it underfits the true physical interactions.

In contrast, the fine-tuned models display a more pronounced overfitting behavior, characterized by a large gap between their low training RMSEs and higher validation RMSEs. For instance, the FT-MultiT model shows a 142% increase in force RMSE from the training (20.3 meV/Å) to the validation set (49.1 meV/Å). This aggressive fitting is expected when fine-tuning on smaller, more specialized datasets. The high standard deviation in the FT-600K model's training RMSE (11.8 meV/Å) also suggests a sensitivity to initialization during the training process.

The fine-tuned models show lower errors than the scratch model on the current near-equilibrium test partition. Table S2 reports the grouped-split values used for the leakage-controlled comparison. RMSE is a pointwise baseline; the barrier and transport analyses below measure process fidelity.

2.2 Detailed Structural and Thermodynamic Analysis

The RDF analysis reveals that all MACE models, regardless of training strategy, successfully reproduce the key structural features of the Cr-doped Sb_2Te_3 system when compared to the AIMD ground truth (Fig. S16b).

All models correctly capture the primary coordination shells, with the first peak positions for Cr–Cr, Cr–Sb, and Cr–Te pairs occurring at approximately 3.0 Å, 3.2 Å, and 2.8 Å, respectively, in excellent agreement with the AIMD reference. The peak heights and positions remain consistent across all training strategies (Figures S16c–f), indicating that the local atomic structure is well preserved regardless of whether the model was trained from scratch, used as a foundation model, or fine-tuned with temperature-specific data.

The primary observable difference between models lies in the smoothness of the RDF curves rather than their fundamental features. The scratch-trained model (Fig. S16d) exhibits slightly smoother RDF profiles compared to the fine-tuned variants (Figures S16e–f). This difference arises from practical computational considerations rather than fundamental accuracy: the scratch model, being more compact with fewer parameters, allowed for longer MD trajectories within the same computational budget, resulting in better statistical sampling. In contrast, the fine-tuned models, while maintaining larger parameter counts from their foundation architectures, required more computational resources per MD step, limiting the total simulation time and resulting in slightly noisier RDF profiles.

Notably, both fine-tuning strategies—whether using single-temperature (600 K) or multi-temperature AIMD data—yield RDF profiles that are nearly indistinguishable for this Cr-doped Sb_2Te_3 system. Within the scope of this test case, this similarity suggests that the structural descriptors learned by the models are not strongly sensitive to the temperature range of the fine-tuning dataset. While this observation is system-specific, it indicates that, for the structural properties examined here, the different training strategies converge to comparable representations of the local atomic environment. The consistent reproduction of key RDF features across all models supports the conclusion that each MLIP captures the essential bonding motifs relevant to this particular material.

2.3 Thermodynamic Ensemble Stability

A detailed analysis of the pressure profiles reveals subtle but important differences between models. The zero-shot MACE Foundation model equilibrates to a slightly different average pressure than the models trained or fine-tuned on our in-house AIMD data. This is likely a consequence of the foundation model being trained on a vast dataset of materials at their 0K equilibrium volumes. A minor mismatch between the foundation model's predicted equilibrium lattice parameters and the true DFT values for our specific Cr:Sb_2Te_3 system at 600K manifests as a persistent non-zero average pressure in an NVT simulation. The fine-tuned and scratch models, having been trained explicitly on data from this system's true ensemble, do not exhibit this deviation.

2.4 Transport Property Analysis

The original FT-600K trajectory is shorter and uses a different sampling protocol from the comparison runs. Its diffusivity and thermal-conductivity values are retained for provenance but excluded from cross-model ranking; under the current trajectories, they are best estimates rather than converged coefficients. The comparison uses matched trajectory length, sampling interval, and block uncertainty before supporting a model-ranking conclusion.

Table S6: **Audited molecular-dynamics transport metrics.** Values are computed from the Revision local re-analysis with block uncertainties. The FT-600K row is shown for provenance but is excluded from cross-model ranking because its trajectory is shorter and sampled differently from the corrected matched protocol analysis.

Model	D (cm^2/s)	R^2 (MSD fit)	κ (W m^{-1} K^{-1})	Status
Foundation	$(0.7 \pm 7.9) \times 10^{-9}$	0.01	$(2.4 \pm 8.6) \times 10^{-3}$	VALID
Scratch	$(-3.4 \pm 3.2) \times 10^{-9}$	0.00	$(0.9 \pm 1.3) \times 10^{-2}$	VALID
FT-600K†	$(3.9 \pm 4.0) \times 10^{-8}$	0.90	$(0.9 \pm 1.2) \times 10^{-2}$	**[INVALID-pending-JB-6]**
FT-MultiT	$(4.8 \pm 4.8) \times 10^{-8}$	0.86	$(2.0 \pm 1.7) \times 10^{-2}$	VALID

†FT-600K 600 K run used steps=2000, interval=5 (`mace_md_eval_0814.py`:359–360); quarantined pending JB-6.
D: block-averaged MSD slope (6 blocks, fit window 5–200 ps). κ: Green–Kubo running integral averaged over 100–200 ps (6 blocks).

2.5 Molecular Dynamics Simulations

2.5.1 Simulation Protocol

MD simulations were performed using the Atomic Simulation Environment (ASE) with the following protocol:

- **Integrator**: Langevin dynamics for NVT ensemble or Nosé-Hoover for NPT ensemble
- **Timestep**: 1.0 fs
- **Friction coefficient**: $\gamma = 0.01$ fs^{-1} (Langevin)
- **Barostat parameters**: $\tau_p = 1000$ fs, $P = 1$ bar (NPT only)
- **Equilibration**: 50,000 steps (50 ps)
- **Production**: 200,000 steps (200 ps) for property calculations
- **Sampling interval**: Every 100 steps for analysis
- **System size**: 20×20×1 supercell (4000 atoms)

Simulations were conducted at two temperatures: 300 K and 600 K, to assess model performance under different thermal conditions. Initial velocities were assigned according to the Maxwell-Boltzmann distribution with removal of center-of-mass motion and angular momentum.

2.6 Property Calculations

2.6.1 Structural Properties

The radial distribution function (RDF) $g(r)$ was computed for all unique atom pairs:

$$g_{\alpha\beta}(r) = \frac{V}{4\pi r^2 \Delta r N_\alpha N_\beta} \left\langle \sum_{i \in \alpha} \sum_{j \in \beta} \delta(r - r_{ij}) \right\rangle \tag{1}$$

where α and β denote atomic species, V is the system volume, N_α is the number of atoms of species α, and the angle brackets denote time averaging.

2.6.2 Dynamic Properties

The mean square displacement (MSD) was calculated for each atomic species:

$$\mathrm{MSD}_\alpha(t) = \left\langle \frac{1}{N_\alpha} \sum_{i \in \alpha} |\mathbf{r}_i(t) - \mathbf{r}_i(0)|^2 \right\rangle \tag{2}$$

Diffusion coefficients were extracted from the linear regime of MSD using the Einstein relation:

$$D_\alpha = \lim_{t \to \infty} \frac{\mathrm{MSD}_\alpha(t)}{6t} \tag{3}$$

The velocity autocorrelation function (VACF) was computed as:

$$C_v(t) = \frac{\langle \mathbf{v}(t) \cdot \mathbf{v}(0) \rangle}{\langle \mathbf{v}(0) \cdot \mathbf{v}(0) \rangle} \tag{4}$$

2.6.3 Thermodynamic Properties

Average potential energy per atom, temperature, and volume (for NPT simulations) were calculated over the production phase:

$$\langle E \rangle = \frac{1}{N_{\text{frames}}} \sum_{i=1}^{N_{\text{frames}}} \frac{E_{\text{pot}}^{(i)}}{N_{\text{atoms}}} \tag{5}$$

with corresponding standard deviations to assess thermal fluctuations.

2.6.4 Transport Properties

For thermal conductivity calculations, the heat flux vector was computed:

$$\mathbf{J} = \frac{1}{V} \left[\sum_i e_i \mathbf{v}_i + \frac{1}{2} \sum_{i \neq j} (\mathbf{F}_{ij} \cdot \mathbf{v}_j) \mathbf{r}_{ij} \right] \tag{6}$$

where e_i is the per-atom energy, $\mathbf{v}_i$ is the velocity, $\mathbf{F}_{ij}$ is the force between atoms i and j, and $\mathbf{r}_{ij}$ is their separation vector. The heat flux autocorrelation function (HFACF) was then computed for subsequent Green-Kubo analysis.

2.7 Representation Learning Feature Extraction

The following physically interpretable descriptors were extracted from MD trajectories:

1. **Energy landscape**: Total potential energy per atom
2. **Force fields**: 3N-dimensional force vectors for all atoms
3. **Structural order parameters**: Radial distribution histograms computed with 100 bins up to 5.0 Å cutoff
4. **SOAP descriptors**: Smooth Overlap of Atomic Positions with:
   - Cutoff radius: 5.0 Å
   - Number of radial basis functions: 8

- Maximum angular momentum: 4
- Gaussian width: 0.3 Å

These features were concatenated into a single vector per configuration, normalized, and projected using t-SNE (perplexity=30, learning rate=200) for visualization. The feature matrix excludes an earlier always-zero placeholder channel intended for a future force-sensitivity descriptor; this channel contained no physical information and is not included in the feature dimension reported for the original-space silhouette calculation. The sensitivity audit additionally evaluates t-SNE perplexities 5, 30, and 50, nominal seeds 0–2, and PHATE seeds 0–2. A finite-difference force-sensitivity descriptor is left as future work rather than used in the present benchmark.

2.8 Evaluation Metrics Definitions

- **Force MAE**: $\frac{1}{3N}\sum_{i=1}^{N}\sum_{\alpha=x,y,z}|F_{i,\alpha}^{\mathrm{MLIP}} - F_{i,\alpha}^{\mathrm{DFT}}|$
- **Energy MAE**: $\frac{1}{N}|E^{\mathrm{MLIP}} - E^{\mathrm{DFT}}|$
- **RMSD Growth Rate**: Linear fit slope of $\mathrm{RMSD}(t) = \sqrt{\frac{1}{N}\sum_{i=1}^{N}|\mathbf{r}_i(t) - \mathbf{r}_i(0)|^2}$
- **Silhouette Score**: Average of $\frac{b_i - a_i}{\max(a_i, b_i)}$ where a_i is mean intra-cluster distance and b_i is mean nearest-cluster distance

2.9 Detailed Migration Pathway Analysis

Table S7: **Audited DFT NEB barrier inventory.** Barriers are forward activation energies relative to the first image of each path. Unconverged and diverged paths are retained for provenance but are not used as final quantitative reference values until restarted calculations satisfy the stated convergence criterion.

Path	N_{img}	$E_{\mathrm{act}}^{\rightarrow}$ (eV)	$E_{\mathrm{act}}^{\leftarrow}$ (eV)	E_{m} (eV)	Status
1-2	5	2.79	1.02	2.79	UNCONVERGED
1-3	5	4.00	3.95	4.00	UNCONVERGED
1-4	5	0.34	0.02	0.34	UNCONVERGED
1-5	5	1.10	0.75	1.10	UNCONVERGED
1-6	5	1.03	0.21	1.03	UNCONVERGED
1-7	5	0.58	0.03	0.58	UNCONVERGED
1-8	5	1.76	0.99	1.76	CONVERGED
1-5_sep2-1	10	7.55	8.60	7.55	UNCONVERGED
1-5_sep2-2	10	800.56	797.73	800.56	DIVERGED
neb_1	10	1.04	1.03	1.04	CONVERGED
neb_2	10	1.56	2.11	1.56	CONVERGED
neb_3	10	1.69	1.67	1.69	CONVERGED
neb_4	10	2.06	1.54	2.06	CONVERGED
neb_5	10	1.42	1.07	1.42	CONVERGED

The grouped low-data control is also retained as a path-resolved diagnostic rather than reduced to a single pooled score. Table S9 reports the three-initialization absolute

Table S8: **Audited Cr migration path-family classification.** Labels are kept separate from canonical cross-system proxy categories. The classification is used to prevent unsupported pooling of model errors; it does not promote any provisional DFT profile to a converged barrier.

System	Audit ID (SI only)	Pathway class	Quantitative use in this manuscript
61 atom	1–2, 1–3	gap-to-QL/into-QL candidate	withheld from pooled ranking
61 atom	1–4, 1–7	within-gap candidate	controlled probe uses one case
61 atom	1–5	gap-to-QL/into-QL; geometry recheck	withheld from pooled ranking
61 atom	1–6	mixed transition; ambiguous	withheld from pooled ranking
61 atom	1–8	probable within-QL candidate; auxiliary	withheld from pooled ranking
81 atom	`neb_1`–`neb_5`	within-QL chain	diagnostic only; DFT gate open

fixed-path errors for the completed model families on the audited 61-atom cases. This table makes the available evidence visible while preserving the distinction between the controlled within-gap comparison and unresolved gap-to-QL/into-QL extrapolation; it is not a final cross-class ranking.

The 61-atom fixed-image evaluations are finite, but they remain single-point model evaluations on historical or candidate DFT images. The independent audit reports residual and SCF-quality limitations, so these profiles are used to identify pathway-dependent failure modes rather than to claim converged DFT barriers. In particular, "gap-to-QL/into-QL" denotes motion from the vdW gap toward or into the interior of a covalently bonded QL, whereas "within-gap" denotes motion remaining in the vdW gap; a mixed or ambiguous case is not forced into either category. Path tortuosity describes the winding geometry of a trajectory, whereas lattice distortion describes the displacement of the surrounding QL; the within-QL label is based on the latter structural event, not on tortuosity alone.

2.9.1 In-Gap Diffusion

This controlled within-gap probe provides a test of fixed-path energetic fidelity. Under the grouped split, train–test proximity audit, and three-initialization retraining (Table S2), FT-600K (OMAT-FT start) has the lowest fixed-path barrier error (0.47 ± 0.12 eV), followed by Scratch-5% (0.68 ± 0.20 eV) and Scratch (1.20 ± 0.27 eV); the historical single-run frame-level estimate of 0.16 eV is not reproduced under this grouped, multi-run protocol. The current Scratch-5% control contains 89 grouped training frames, not a literal 1000-frame subset, so it supports a low-data comparison but does not isolate a pure pretraining effect at exactly matched data volume. FT-Multi_T

Table S9: **Descriptive class-resolved fixed-path errors for the grouped low-data control.** Values are mean ± sample standard deviation across three independent training initializations for FT-600K, Scratch-5%, and full Scratch. The audit ID is shown only to connect this SI table to the raw evidence; the narrative uses the pathway classes in the second column. The controlled within-gap probe is the primary A5 comparison. The other rows are retained as diagnostics because their pathway classification, geometry audit, and/or DFT reference quality are not yet uniformly closed; they must not be interpreted as a pooled higher-distortion ranking.

Audit ID (SI only)	Pathway class	DFT candidate	FT-600K	Scratch-5%	Scratch
1-2	gap-to-QL/into-QL candidate	2.79	9.60 ± 0.57	0.33 ± 0.31	0.51 ± 0.11
1-3	gap-to-QL/into-QL candidate	4.00	10.40 ± 0.44	1.22 ± 0.66	0.72 ± 0.55
1-4	within-gap controlled probe	0.34	0.47 ± 0.12	0.68 ± 0.20	1.20 ± 0.27
1-5	gap-to-QL/into-QL; geometry recheck	1.10	9.70 ± 0.20	1.07 ± 0.31	1.77 ± 1.00
1-7	additional within-gap candidate	0.58	0.21 ± 0.18	0.25 ± 0.21	0.33 ± 0.06

The DFT candidate barriers are provisional anchors for this audit, not final converged references. Full-precision values and row-level provenance are retained in `grouped_neb_barrier_summary.csv`; manuscript-facing values are rounded to two decimals.

was not part of this grouped-retraining round. The ± values quantify model-level initialization variation, while pathway-level variation remains unresolved across the audited family.

For a matched foundation-model comparison, we evaluated ten single-checkpoint models on the same five fixed images. The four main representatives (MP-0b3, MPA0, OMAT-0 medium, and MH-1) produced barriers of 1.50, 1.79, 1.60, and 1.84 eV, respectively, against the provisional DFT reference of 0.34 eV. The ten individual profiles are shown in Fig. S19a and the family means with pointwise ±1 standard-deviation bands are shown in Fig. S19b. The family bands describe variation among sub-models within each foundation family, not training-initialization uncertainty. The profiles are qualitatively smooth but systematically elevated relative to the DFT reference, reinforcing that equilibrium accuracy and fixed-path migration fidelity are distinct observables.

2.9.2 Higher-Distortion and Within-QL Candidates

Within-QL migration is defined here as Cr entering the interior of a covalently bonded QL, which can require substantial lattice distortion. The audit does not support pooling all higher-distortion candidates into one quantitative OOD score. The audit-ID table records which cases are gap-to-QL/into-QL candidates, which require a geometry recheck, and which are mixed or auxiliary; their historical or candidate DFT references also remain subject to the stated quality gates.

The higher-distortion profiles vary across models, but the current audit does not identify a defensible Scratch ranking. Grouped multi-seed retraining supplies model-level uncertainty for the completed in-gap comparison; path-level classification and DFT-convergence uncertainty remain open for this OOD family.

2.9.3 Cross-element diagnostic and scope of generalizability

As a scope audit, three Cr-trained broad-roster checkpoints from the recorded final epoch were evaluated on archived Cu/Li/Na paths in the same Sb_2Te_3 host. They matched exactly across the recorded 90 state-dict tensors and gave validation energy RMSEs of 11.33, 12.16, and 12.59 meV/atom and force RMSEs of 203.62, 208.76, and 222.02 meV/Å. Across ten archived paths, all image energies and forces were finite and image-count, endpoint-identity, and PBC checks passed; 8/10 passed the package's < 0.10 eV screening threshold. All ten DFT path records nevertheless retain SCF-convergence warnings and fail the scientific DFT gate. The screening threshold is therefore not interpreted as a barrier-accuracy or transfer-validation result.

For the six gap-like paths, the maximum relative-energy diagnostics across the three checkpoints are 0.06 ± 0.01 eV for Cu gap-diffuse, 0.02 ± 0.01 eV for Cu gap2QL, 0.10 ± 0.04 eV for Li gap-diffuse, 0.07 ± 0.03 eV for Li gap2QL, 0.30 ± 0.17 eV for Na gap-diffuse, and 0.12 ± 0.04 eV for Na gap2QL. These are diagnostic snapshot variations, not DFT barrier-error uncertainties. Broader cross-element transfer remains a validation target rather than a completed generalization claim.

Table S10: **A4 cross-element diagnostic.** The Cr-trained broad-roster models were evaluated on archived Cu/Li/Na paths in the same host. All ten DFT records retain SCF-convergence warnings and therefore fail the scientific DFT gate. The final-column diagnostic is the mean ± sample SD of the maximum MLIP relative energy across three checkpoints; it is not a DFT barrier error.

Element	Path	DFT gate	< 0.1 eV screen	Max relative energy (eV)	Seed range (eV)
Cu	Cu_neb	failed	no	123.43 ± 3.95	8.45
Cu	Cu_neb-QLs	failed	yes	1.15 ± 0.03	0.08
Cu	Cu_neb-gap-diffuse	failed	yes	0.06 ± 0.01	0.02
Cu	Cu_neb-gap2QL	failed	yes	0.02 ± 0.01	0.03
Li	Li_neb-QLs	failed	yes	1.14 ± 0.02	0.05
Li	Li_neb-gap-diffuse	failed	yes	0.10 ± 0.04	0.09
Li	Li_neb-gap2Ql	failed	yes	0.07 ± 0.03	0.06
Na	Na_neb-QLs	failed	no	1.62 ± 0.10	0.25
Na	Na_neb-gap-diffuse	failed	yes	0.30 ± 0.17	0.39
Na	Na_neb-gap2QL	failed	yes	0.12 ± 0.04	0.08

2.9.4 Implications for MLIP Development

These findings highlight critical considerations for practical MLIP deployment:

- **Validation Insufficiency**: Testing only on stable configurations masks critical failures at transition states

- **Hidden Extrapolation**: Models can appear accurate at trajectory endpoints while failing at crucial transition states
- **Data Quality over Quantity**: Task-specific training data outweighs larger but less relevant datasets
- **Foundation Model Limitations**: Pre-trained models require careful adaptation for processes involving significant atomic rearrangement

The results motivate advanced strategies like uncertainty-guided active learning that can identify high-uncertainty regions and intelligently augment training sets for improved extrapolative performance.

2.10 Nudged Elastic Band (NEB) Benchmark for Cr Migration

To directly assess the practical performance of each model on a physically critical task, we performed a Nudged Elastic Band (NEB) calculation for a Cr atom migration event, as shown in Figure S25. This calculation serves as an effective and challenging probe to distinguish the models' stability and predictive accuracy when exploring unseen transition-state geometries. The results reveal a dramatic difference in performance that is not apparent from the RMSE metrics alone.

Both the Scratch and fine-tuned models (FT-600K and FT-MultiT) show a rapid increase in the maximum image force (f_{max}) during the recorded NEB optimization, and the runs terminate before a converged band is obtained. These runs establish instability under the recorded self-relaxed NEB protocol for this test; the microscopic cause is not resolved by the present data, and the records are not used as a same-system barrier ranking.

The historical SI Fig. S3 record reported a 0.41 eV Foundation value, but the Revision-1 audit found that it does not correspond to the main-text 61-atom within-gap system: it uses an 82-atom, two-Cr system with twelve images rather than the 61-atom, one-Cr, five-image path used for the fixed-geometry comparison. The historical calculation also used inconsistent endpoint and movable-image calculators, and the final band contains a serialization defect. Uniform rescoring of the preserved band gives a forward endpoint-referenced barrier of 0.00 eV, a reverse value of 0.38 eV, and a whole-band span of 1.04 eV, not 0.41 eV. We therefore mark the historical 0.41 eV value `EXCLUDE_FROM_PUBLICATION` and retain it only to document the source of the original discrepancy. A valid same-system self-consistent MLIP-NEB comparison remains a future validation target.

2.11 Audited Representation and Surrogate Diagnostics

The representation audit uses 404 trajectory frames and 6,191 descriptors after removing the asserted all-zero force-sensitivity stub. In the original descriptor space, the overall Euclidean silhouette is 0.33 and the cosine silhouette is 0.28. After z-scoring each descriptor, the Euclidean silhouette is -0.02; a 95% variance PCA of the z-scored matrix gives -0.03 in 364 dimensions. Because the descriptor vector combines total energy, forces, RDF features, and SOAP components, the raw-space value is scale-sensitive and is interpreted as descriptive separation of the recorded model outputs, not as evidence for disjoint local-geometry manifolds.

The projection audit gives a t-SNE silhouette of 0.40 ± 0.03 across perplexities 5, 30, and 50, with a range of 0.36–0.42 and trustworthiness of 0.963–0.973. PHATE gives 0.47 ± 0.00 across three nominal seeds. At fixed perplexity, the nominal t-SNE seeds are bitwise-identical under the locked PCA initialization; the reported spread therefore reflects perplexity sensitivity and is not treated as independent seed uncertainty. The projections are retained as qualitative diagnostics consistent with different descriptor distributions, not as direct mechanistic explanations.

For the SHAP analysis, the corrected frame-aligned workflow uses 150 configurations, 390 SOAP descriptors, and 17 structural descriptors. The 5-fold cross-validated surrogate R^2 values are 0.98 for FT-600K, 0.88 for FT-MultiT, and 0.97 for Scratch, with corresponding MAEs of 0.04, 0.03, and 0.03 eV. The largest corrected FT-MultiT feature is `Cr-Sb_n43_l0`; Cr–Cr descriptors occur in some rankings but are not a stable dominant channel across models. These values describe the surrogate's sensitivity to predictors of absolute energy error and do not establish a causal attribution within MACE.

The complementary endpoint campaign contains 846 fixed-cell native-MACE FIRE relaxations across six Cr/V/Mn–Sb_2Te_3/Bi_2Te_3 systems and three perturbation arms; all tasks pass the recorded scientific gates. It is reported in the endpoint appendix as a local-stability probe. It is not an isolated Cr–Cr SOAP-channel perturbation, not a DFT or magnetic validation, and not a causal test of the SHAP ranking; the exact isolated Cr–Cr perturbation remains a validation gap.

Table S11: **Audited separation metrics across projected and descriptor spaces.** Projected values are descriptive baselines. The raw descriptor-space value is scale-sensitive because the vector mixes energy, force, RDF, and SOAP components; the z-scored and PCA checks are included to expose that sensitivity.

Space	Overall silhouette	Interpretation
Original 6,191-d, Euclidean	0.33	Raw descriptor-space baseline; scale-sensitive
Original 6,191-d, cosine	0.28	Alternative raw-space metric
Original 6,191-d, z-scored Euclidean	-0.02	Scale-controlled overlap check
PCA 95%, z-scored Euclidean	-0.03	364-dimensional conservative check
t-SNE, $p \in \{5, 30, 50\}$, 3 nominal seeds	0.40 ± 0.03	Range 0.36–0.42; trustworthiness 0.963–0.973
PHATE 2D, 3 nominal seeds	0.47 ± 0.00	Range 0.47265–0.47277

3 Six-system multi-arm endpoint perturbation campaign

We evaluated the response of six dopant–host systems, Cr/V/Mn embedded in Sb_2Te_3 and Bi_2Te_3, to three endpoint-relaxation arms: the historical unperturbed endpoint (Base), a +0.20 Å dopant displacement (Perturb A), and a displacement of

Table S12: **Audited SHAP surrogate fidelity.** The previous in-sample R^2 values are superseded by 5-fold cross-validation on the corrected 150-frame, frame-aligned surrogate workflow. The table reports surrogate predictive fidelity, not causal fidelity of MACE.

Model	n	5-fold CV R^2	5-fold CV MAE (eV)
FT-600K	150	0.98	0.04
FT-MultiT	150	0.88	0.03
Scratch	150	0.97	0.03

Table S13: **Measured multi-arm endpoint-relaxation summary.** Values are medians over the 47 historical endpoint sites for each of six host–dopant systems and each perturbation arm. "Conv." reports the number of FIRE relaxations reaching $f_{\max} \leq 0.03$ eV Å^{-1}; E_f is the final MACE energy, $F_{\max}$ the final maximum force, $d_{\max}$ the maximum atomic displacement from the perturbed input, and "steps" the FIRE step count. All 846 formal tasks passed the finite-value, topology, minimum-distance, displacement, and scientific-success gates. The Foundation-MACE model and fixed-cell FIRE protocol are the same across arms; this table reports model-relaxation observables, not DFT or magnetic conclusions.

	Base				**Perturb A:** $+x\,0.20$				**Perturb B:** $(-x, +y, +z)$			
System	**Conv.** (47)	E_f (eV)	$F_{\max}$ (eV/Å)	$d_{\max}$ (Å)	**Conv.** (47)	E_f (eV)	$F_{\max}$ (eV/Å)	$d_{\max}$ (Å)	**Conv.** (47)	E_f (eV)	$F_{\max}$ (eV/Å)	$d_{\max}$ (Å)
Cr@Sb2Te3	47/47	-299.68	0.03	0.02	47/47	-299.69	0.03	0.20	47/47	-299.70	0.03	0.19
V@Sb2Te3	47/47	-299.47	0.03	0.13	47/47	-299.48	0.03	0.23	47/47	-299.46	0.03	0.24
Mn@Sb2Te3	47/47	-299.34	0.03	0.13	47/47	-299.34	0.03	0.23	47/47	-299.34	0.03	0.23
Cr@Bi2Te3	47/47	-300.84	0.03	0.47	47/47	-300.82	0.03	0.49	47/47	-300.83	0.03	0.47
V@Bi2Te3	47/47	-300.59	0.03	0.50	47/47	-300.61	0.03	0.53	47/47	-300.58	0.03	0.55
Mn@Bi2Te3	47/47	-300.71	0.03	0.50	47/47	-300.71	0.03	0.53	47/47	-300.71	0.03	0.53

Arm definitions. Base is the unperturbed historical endpoint. Perturb A displaces the dopant by $(+0.20, 0, 0)$ Å; Perturb B uses $(-0.10, +0.14, +0.10)$ Å(norm 0.20 Å). Medians summarize distributions and should not be read as endpoint-level equivalence.

$(-0.10, +0.14, +0.10)$ Å (Perturb B). The campaign contains 47 endpoint sites per system and arm, for 846 formal MACE relaxation tasks. All tasks passed the finite-value, topology, minimum-distance, displacement, and scientific-success gates. These results are model-relaxation observables from Foundation-MACE with fixed-cell ASE FIRE relaxation; they are not DFT, magnetic, or oxidation-state assignments.

3.1 Interpretation and reproducibility

The complete endpoint-level data used for these figures are provided in `revision1/appendix/data/endpoint_level_summary.csv` and its JSON counterpart. The summary table is generated from

`revision1/appendix/data/multiarms_summary.csv`. The plotting source is preserved as `revision1/appendix/scripts/plot_campaign_atlas.py`; it uses the project Plot Atlas primitives and regenerates the three PDF, PNG, and SVG figures from the supplied data. Overleaf compiles the checked-in PDF assets; the data and script are included alongside them so that the figures remain auditable and reproducible outside the hosted compiler.

The dominant qualitative result is that all three arms are scientifically accepted, while perturbation magnitude and response depend strongly on both host and dopant. In particular, the Bi_2Te_3 systems exhibit larger endpoint displacements than the corresponding Sb_2Te_3 systems, whereas the final force distributions remain tightly concentrated near the FIRE acceptance threshold. These patterns support using endpoint perturbations as a controlled local-stability probe, but do not establish transferability to DFT energetics or to finite-temperature migration barriers.

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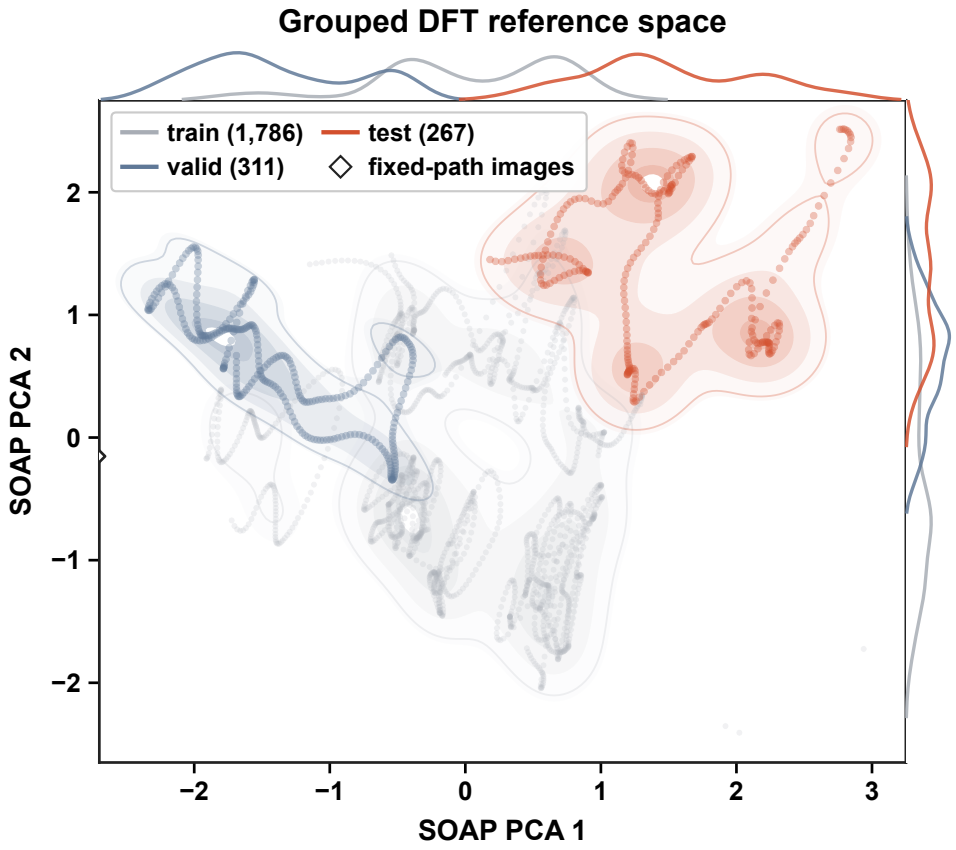


Fig. S2: **Grouped DFT reference space.** SOAP/PCA is fitted on the trajectory-grouped training partition only and applied without refitting to 1,786 training, 311 validation, 267 test, and five fixed-path pathway-1–4 images. The training partition contains 1,783 periodic AIMD frames and three isolated-atom calibration frames; complete source AIMD trajectories are assigned to one partition. Soft two-dimensional density surfaces and calibrated marginal supports show the grouped split, while diamonds identify the fixed-path images. The latter are kept as a separate geometry class because the audited pathway-1–4 images contain 61 atoms, whereas the grouped AIMD cells contain 81–86 atoms. This panel is a descriptive representation diagnostic rather than a scalar OOD score.

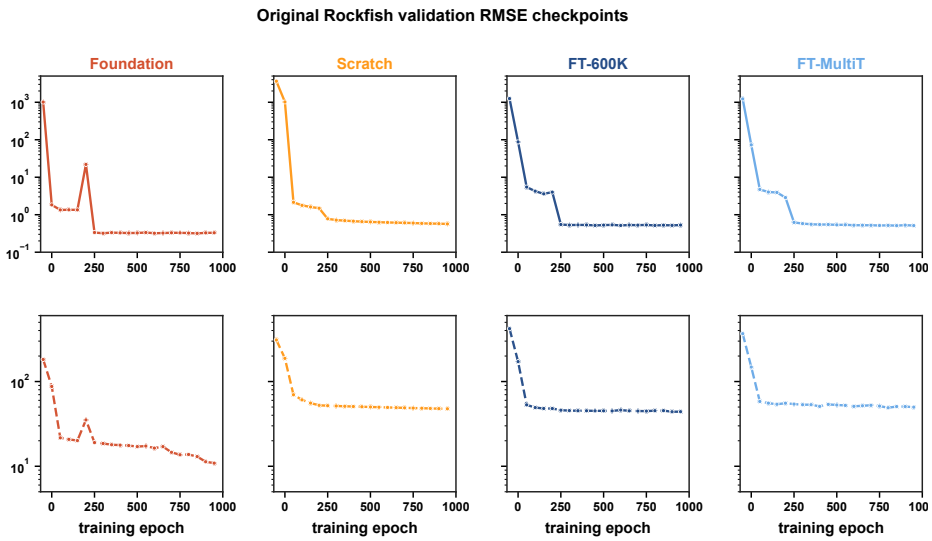


Fig. S3: **Original Rockfish validation RMSE checkpoints.** The four columns follow the manuscript order Foundation, Scratch, FT-600K, and FT-MultiT. The upper row reports energy RMSE per atom and the lower row reports force RMSE, both converted from the original JSONL training logs to meV-based units. Each curve contains the unique default-head `mode=eval` checkpoints from the corresponding run 123 log; duplicate log checkpoints are retained in the raw table but omitted from the plotted trace. These are descriptive single-run validation histories from the original training campaigns, not grouped held-out test errors, seed uncertainty, or a universal model ranking. The source paths, SHA256 values, and model-lineage status are recorded in the accompanying RMSE manifest.

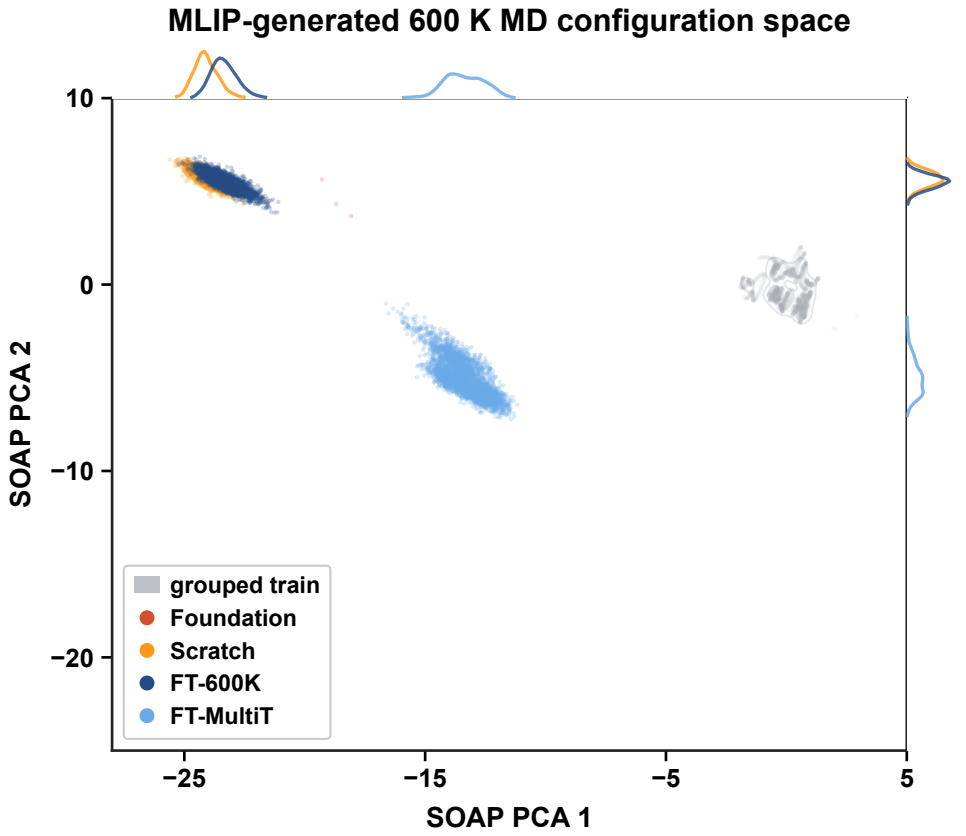


Fig. S4: **MLIP-generated 600 K MD configuration space.** SOAP/PCA distributions from the stored MLIP-generated continuation segments are overlaid on the grouped-training density using the same PCA transform as panel (a). The stored rows are Foundation 4,961 (3 SOAP-valid and 4,958 short-contact skips), Scratch 4,001 (1,310 valid and 2,691 short-contact skips), FT-600K 4,601 valid, and FT-MultiT 4,001 valid. Short-contact rows remain in the raw inventory but are excluded from SOAP after the geometry-safety precheck. These clouds are MLIP-only distributional diagnostics rather than DFT errors; the common MD velocity seed means that the branches are not independent velocity replicas or calibrated uncertainty estimates.

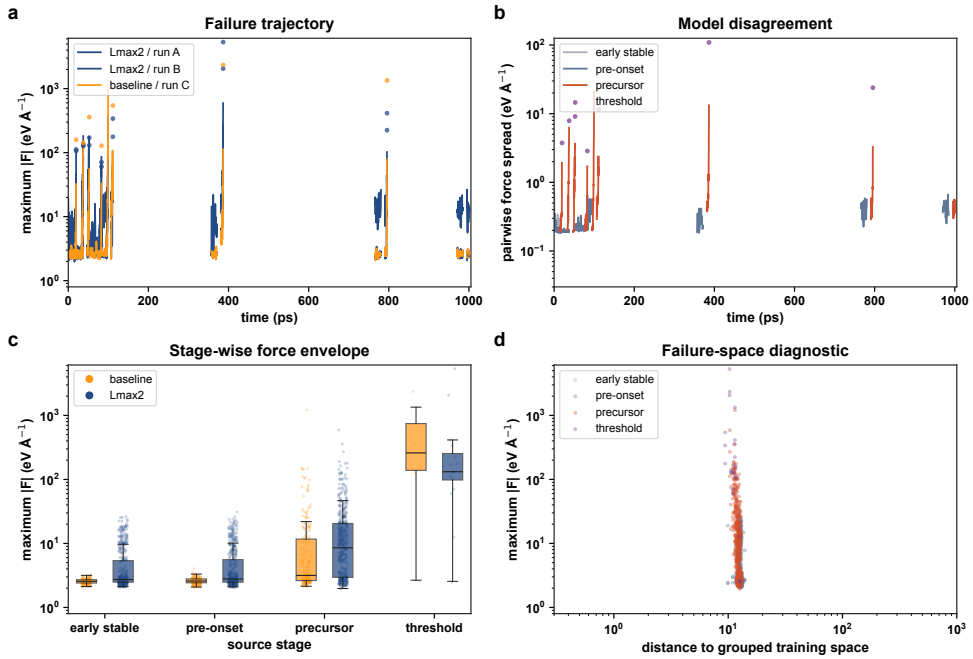


Fig. S5: **Stage-resolved MLIP-only failure/CAR diagnostic.** (a) Maximum force along the 639-candidate failure trajectories for the baseline and $l_{\max} = 2$ branches. (b) Paired model force disagreement across the same stage windows. (c) Stage-wise force envelopes for the two model branches. (d) Maximum force as a function of distance to the grouped training representation. The stage-balanced samples are descriptive diagnostics of failure progression and model disagreement; they are not DFT errors, calibrated uncertainties, or a monotonic scalar OOD ranking.

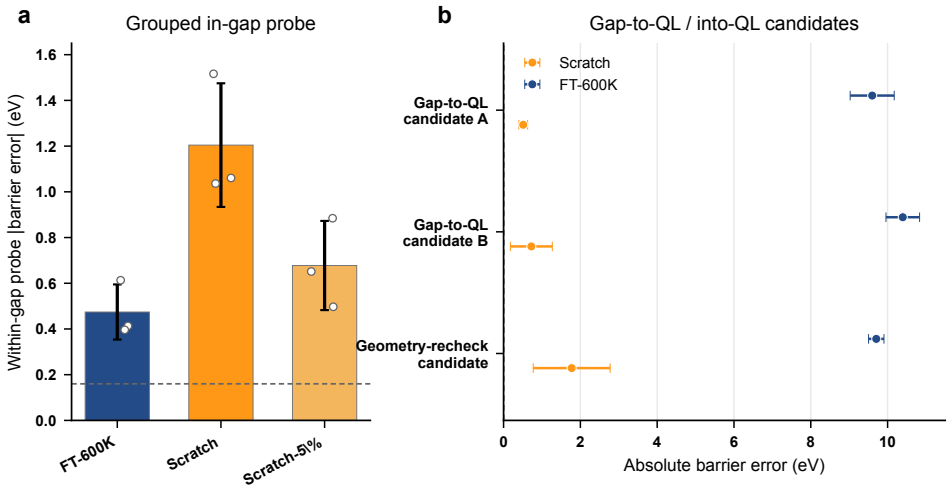


Fig. S6: **Seed-resolved kinetic uncertainty.** The grouped within-gap comparison and representative gap-to-QL/into-QL candidate errors are shown with the individual-run observations and mean ± sample standard deviation. One higher-distortion candidate is comparatively stable for Scratch within the tested initializations, whereas the other candidates are more seed-sensitive. Higher-distortion DFT references remain provisional, and one candidate requires a geometry recheck; the exact audit IDs are retained in the accompanying table and raw evidence.

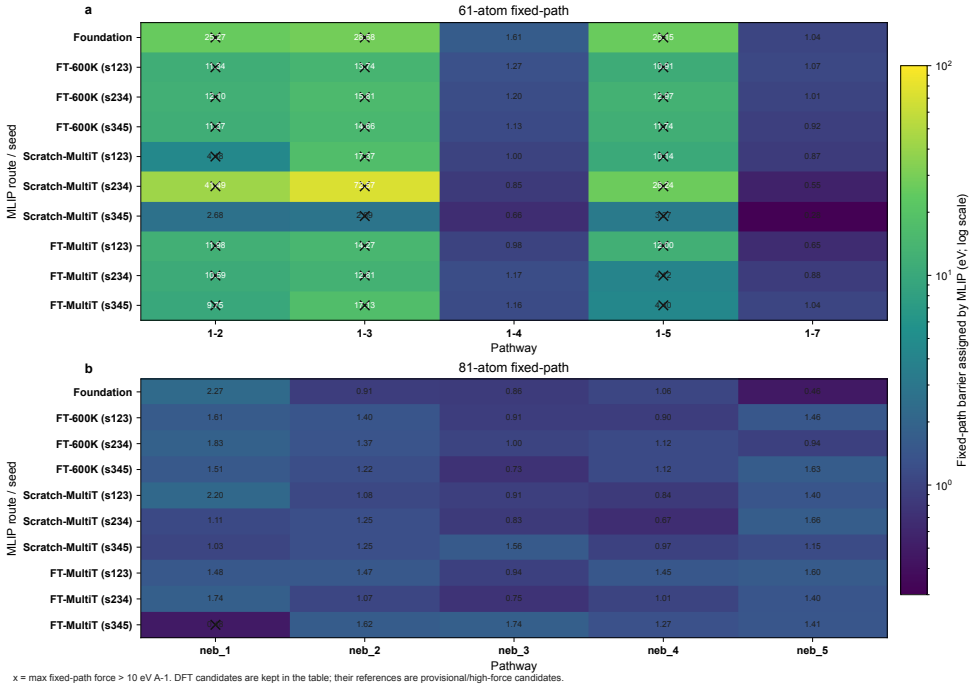


Fig. S7: **Supporting fixed-path NEB protocol matrix.** The 61-atom and 81-atom panels show absolute barrier errors assigned on common DFT-defined image sequences for the recorded model/seed routes. The 61-atom rows include the strict grouped routes and separate Scratch-Multi$_T$ and FT-Multi$_T$ lineages; the 81-atom references are high-force candidate paths. Crosses mark finite fixed-path outputs whose maximum image force exceeds 10 eV/Å. All candidate DFT references and force-pathology cells are retained for audit but are not pooled into a final higher-distortion ranking.

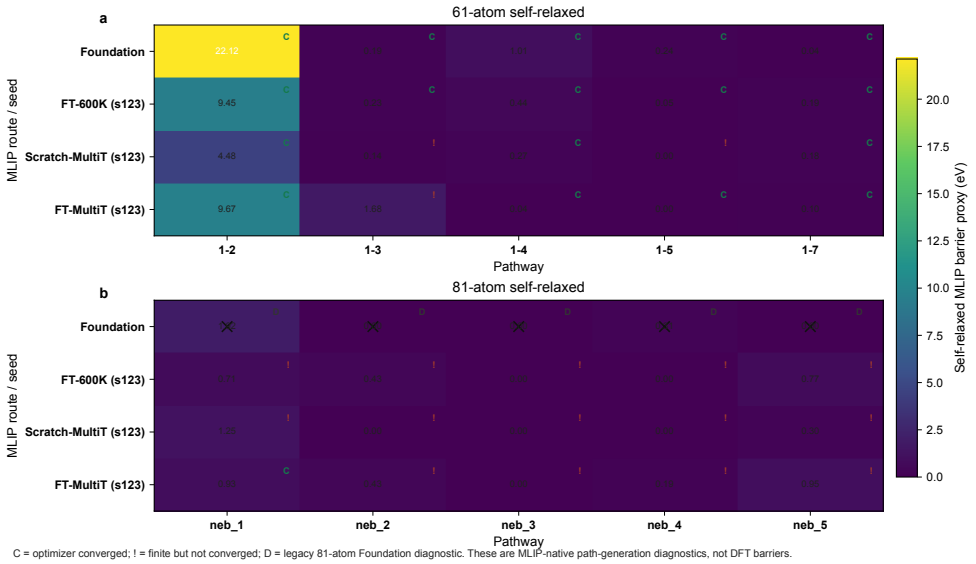


Fig. S8: **Supporting self-relaxed NEB protocol matrix.** Values are MLIP-native barrier proxies after model-in-the-loop path optimization. "C" denotes optimizer convergence under the recorded MLIP protocol, "!" denotes a finite but non-converged run, and "D" denotes a legacy 81-atom Foundation diagnostic with incomplete convergence fields. The relaxed images have not been DFT-rescored; the matrix therefore documents path-generation stability and failure modes, not physical self-relaxed barrier accuracy.

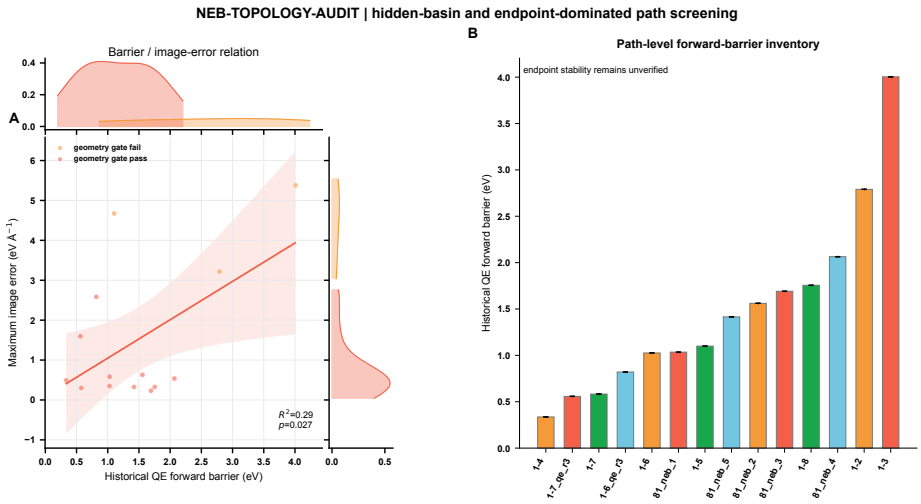


Fig. S9: **Reused path-topology audit supporting the A3 classification boundary.** This historical review-book asset (FIG-211) relates image-error diagnostics to the path inventory and exposes endpoint-dominated or hidden-basin candidates. It is retained as a provenance and pathway-classification audit. The 0.336 eV path-1–4 value is a candidate fixed-path reference rather than an accepted converged barrier; the figure does not provide seed uncertainty, endpoint relaxation, NEB convergence, or DFT rescoring of MLIP-relaxed images.

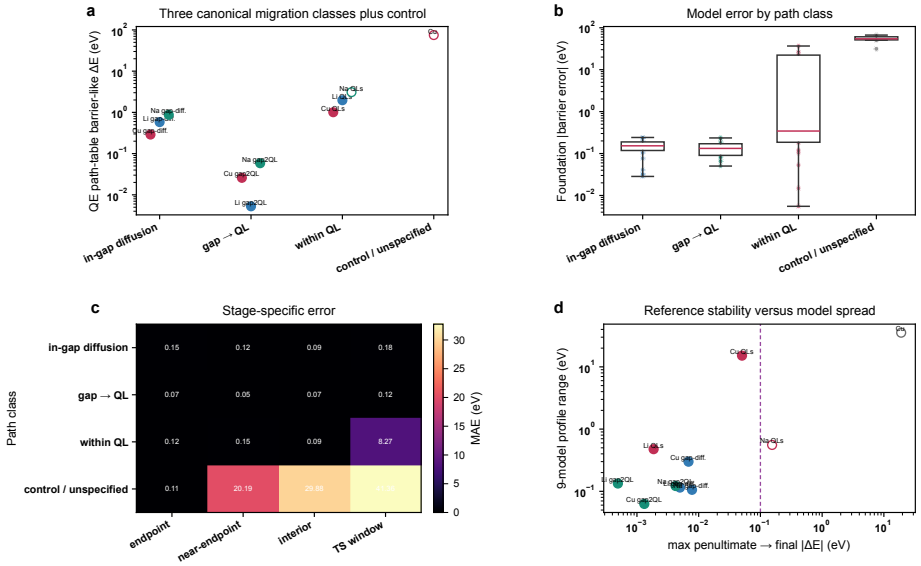
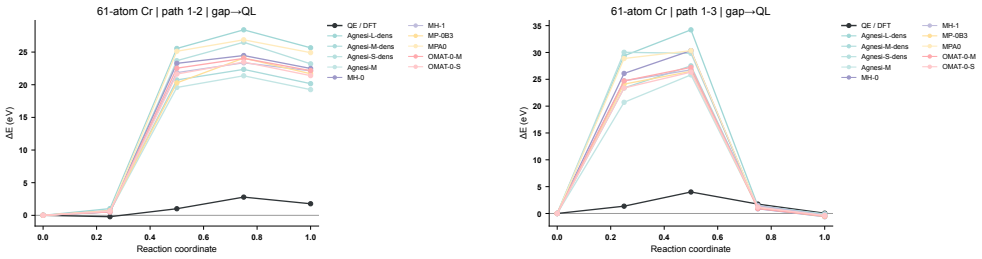


Fig. S10: **Reused path-type summary for the A3 audit.** This historical review-book asset separates the controlled within-gap/interpolative probe from gap-to-QL/into-QL and other higher-distortion candidates. It is retained to make the pathway-class boundary explicit; the classifications and DFT references remain provisional where flagged and are not pooled into a final hard-OOD ranking.



(a) Path 1–2. (b) Path 1–3.

Fig. S11: **Reused higher-distortion path profiles.** These historical fixed-image profiles provide path-resolved context for the deep-penetration candidates. They are descriptive model/path diagnostics rather than the grouped seed-resolved uncertainty results; candidate DFT anchors and pathway labels remain provisional.

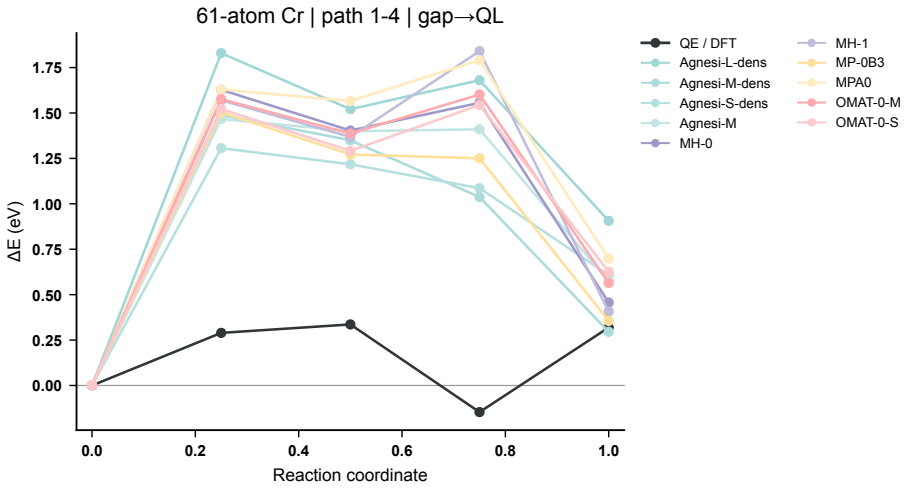
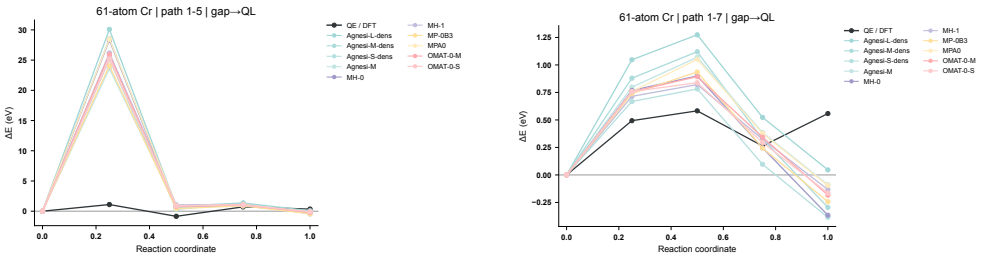


Fig. S12: **Reused controlled path-1–4 profile.** The historical export labels this panel "gap→QL", whereas the current grouped audit treats the path-1–4 record as the controlled within-gap reference; this label reconciliation is retained in the provenance notes. The profile visualizes the common DFT-defined images used for the current comparison. The grouped three-initialization errors in Table S2 and Fig. S6 are the quantitative uncertainty result; this profile is not a self-relaxed NEB result and the DFT reference remains provisional.



(a) Path 1–5; geometry recheck.

(b) Path 1–7; in-gap candidate.

Fig. S13: **Reused geometry/classification boundary profiles.** The path-1–5 panel retains its geometry-recheck status and path 1–7 remains a separate in-gap candidate. These pages support the decision to show path-resolved descriptive evidence without pooling unresolved classes into a hard-OOD score.

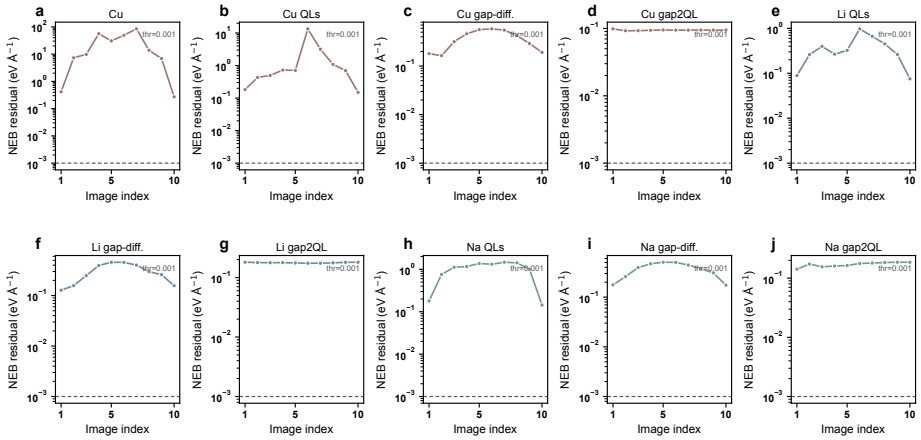


Fig. S14: **Reused NEB residual audit.** The historical audit shows residual-force and convergence limitations in the DFT path records. It supports the provisional-reference boundary used for the higher-distortion rows but is not a converged DFT barrier result.

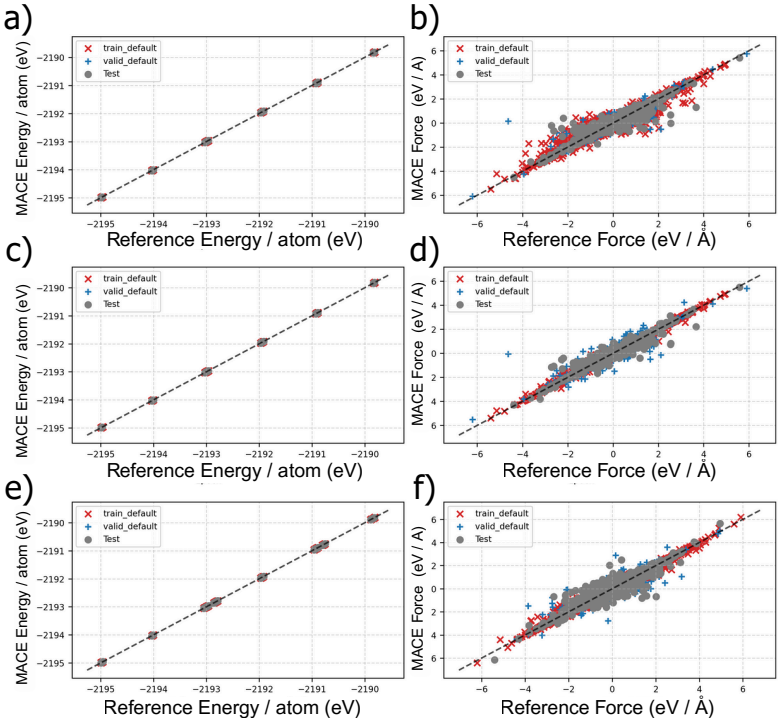


Fig. S15: **Parity plots comparing MACE-predicted energies and forces against DFT reference calculations for the test set.** (a, b) Model trained from scratch, (c, d) fine-tuned 600 K model ("FT-600K"), and (e, f) fine-tuned multi-temperature model ("FT-MultiT"). Parity plots for energies (a, c, e) demonstrate excellent agreement for all models, while force parity plots (b, d, f) show strong overall correlation but subtle differences in error distributions that are not distinguishable visually. These results highlight the necessity of quantitative and property-based benchmarks for robust model evaluation.

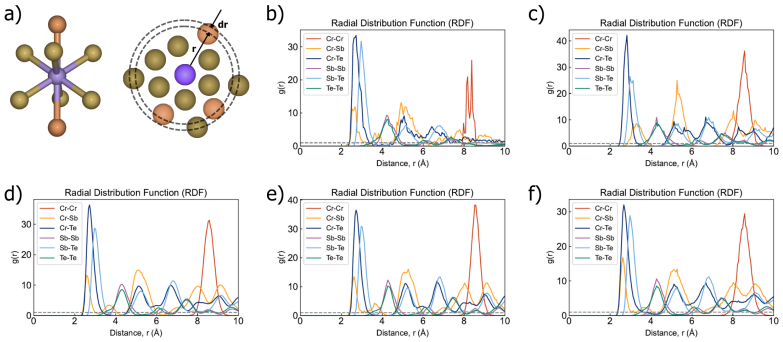


Fig. S16: **Radial distribution function (RDF) analysis of Cr-doped Sb_2Te_3 at 600 K.** (a) Schematic illustration of the local atomic environment around a Cr dopant (purple) in the Sb_2Te_3 lattice, showing the first coordination shell of Te atoms (gold) and second-shell Sb atoms (orange). The dashed circles indicate the radial distances used for RDF calculation. (b-f) Computed RDFs for different atom pairs from molecular dynamics simulations: (b) Ab initio molecular dynamics (AIMD) ground truth reference, (c) MACE foundation model without fine-tuning, (d) MACE model trained from scratch on Cr-Sb_2Te_3 data, (e) MACE model fine-tuned with 600 K AIMD data ("FT-600K"), and (f) MACE model fine-tuned with multi-temperature AIMD data ("FT-MultiT"). All simulations were performed at 600 K with 5×5×1 supercells for 200 ps. The RDF peaks correspond to characteristic interatomic distances in the doped structure, with the first peak representing nearest-neighbor correlations.

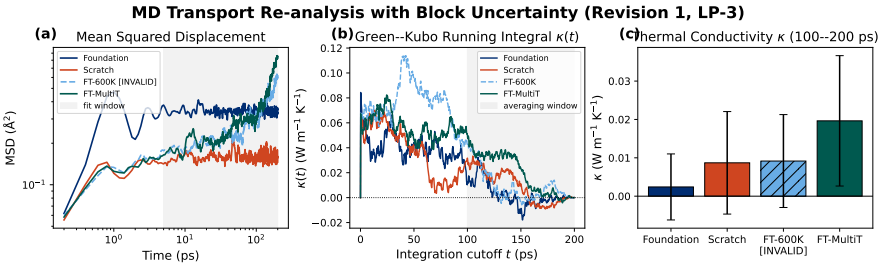


Fig. S17: **Revision 1 transport re-analysis with block uncertainty.** (a) Mean-squared displacement fits used to estimate diffusion coefficients. (b) Green–Kubo running integrals for thermal conductivity. (c) Block-averaged thermal conductivities over the 100–200 ps integration window. The FT-600K value is shown for traceability but is excluded from ranking because its trajectory is shorter and sampled differently from the comparison runs.

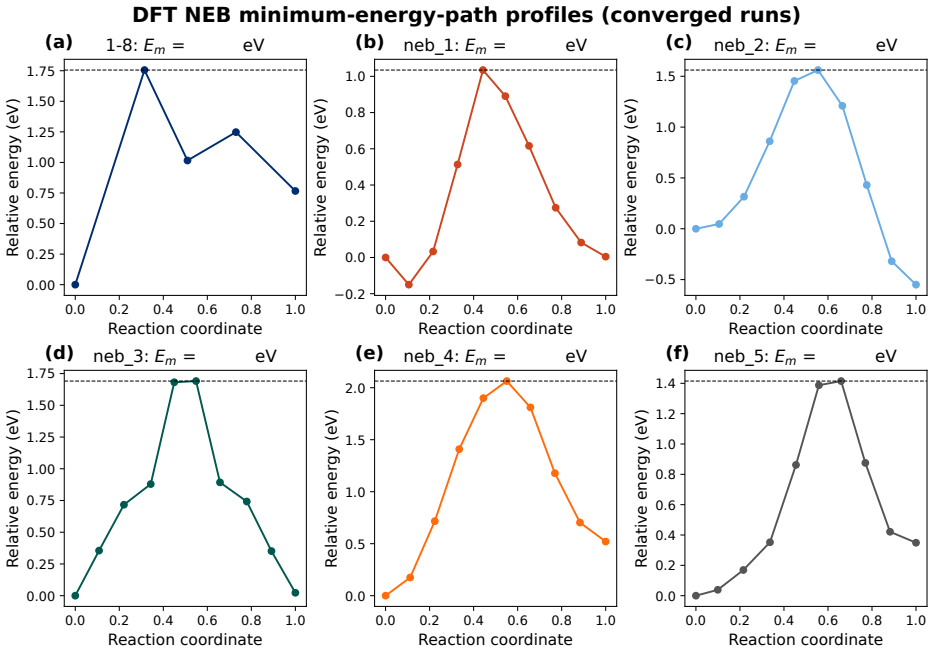


Fig. S18: **Audited DFT NEB minimum-energy-path profiles.** The panels show the currently parsed NEB profiles used to populate Table S7. Candidate or unconverged paths remain flagged in the barrier inventory and source ledger rather than treated as finalized reference data.

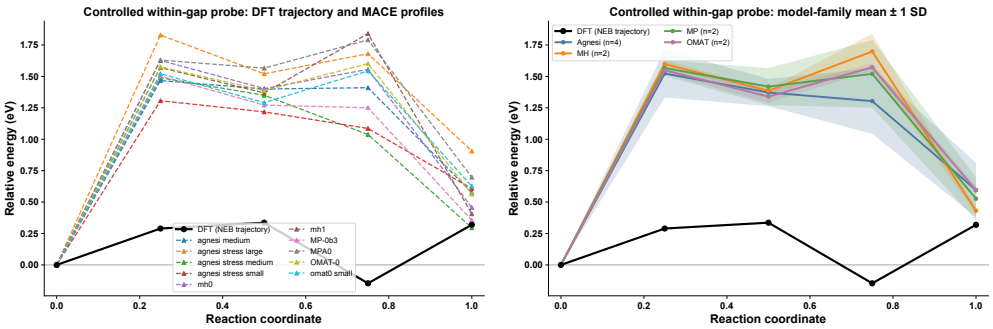


Fig. S19: **Foundation-model fixed-path energy profiles for the controlled within-gap probe.** (a) Relative-energy profiles for all ten foundation checkpoints evaluated on the same five DFT-path images. (b) Family-aggregated profiles for Agnesi, MH, MP, and OMAT; shaded bands denote pointwise ±1 standard deviation across sub-models within each family, not training-initialization uncertainty. The DFT reference is the provisional 0.34 eV barrier candidate. These are fixed-image single-point profiles; no model-relaxed NEB convergence is implied.

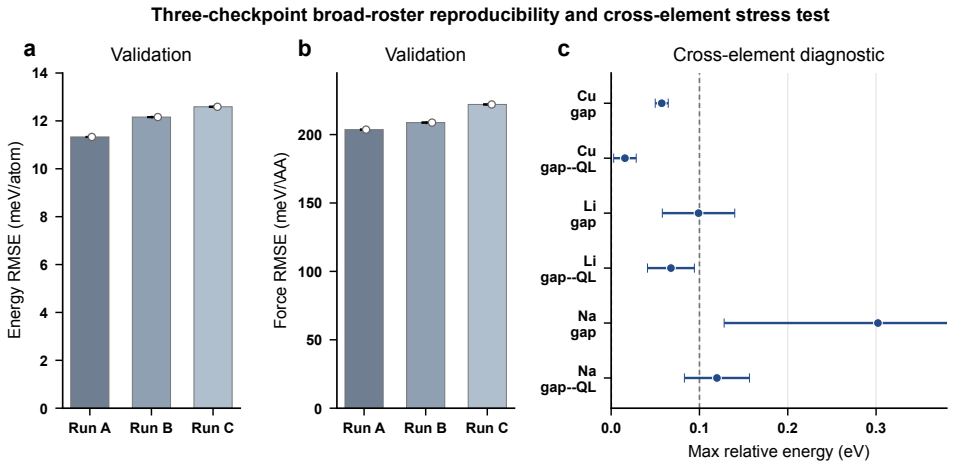


Fig. S20: **Cross-element diagnostic for scope control.** (a,b) Three-seed validation RMSEs for the broad-roster checkpoints. (c) Maximum relative-energy diagnostics for six gap-like Cu/Li/Na paths; the dashed line marks the 0.10 eV screening threshold. All ten archived DFT paths retain SCF-convergence warnings, so the panels are descriptive and do not establish cross-element barrier accuracy.

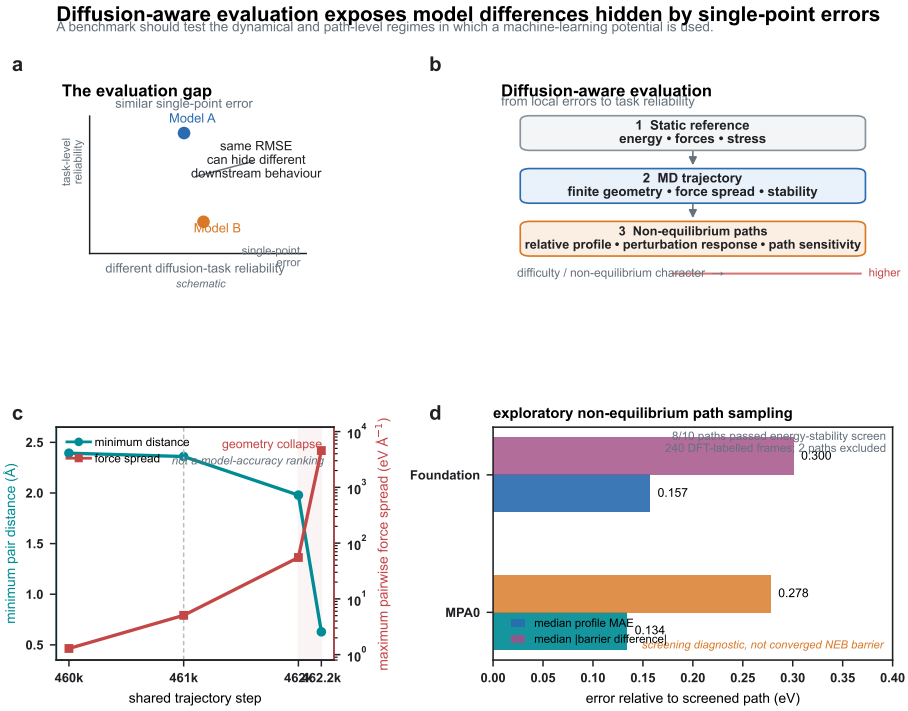


Fig. S21: **Reused diffusion-aware evaluation overview.** This historical review-book asset connects model selection to increasingly demanding dynamic and pathway probes. It is retained as visual context for the candidate task-conditioned framework demonstrated on one Cr-doped Sb_2Te_3 system and does not establish universal transfer.

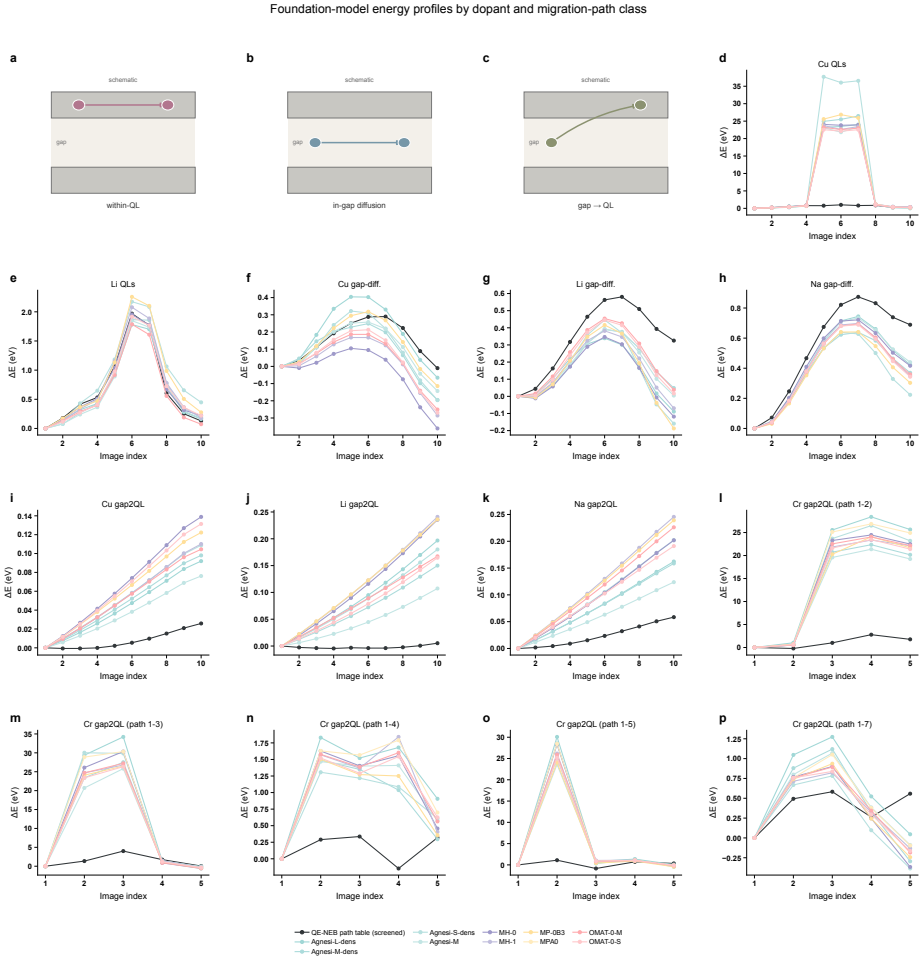
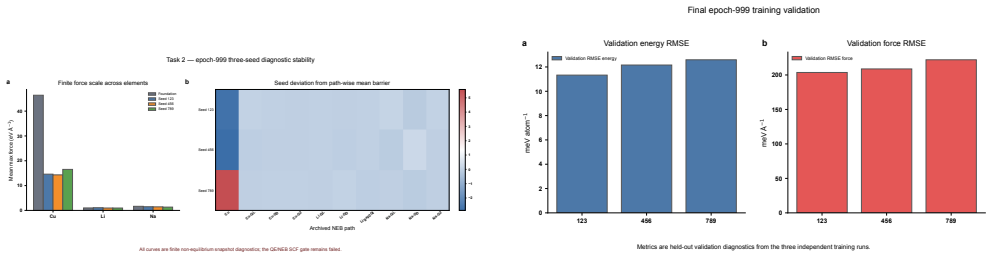


Fig. S22: **Reused task-type matrix.** This historical matrix distinguishes static, equilibrium, fixed-path, self-relaxed, and broader pathway diagnostics. It supports the current evaluation ladder and operator distinction; it is not a universal difficulty scale or model ranking.



(a) Three-seed stability.

(b) Three-seed validation.

Fig. S23: **Reused cross-element diagnostic pages.** The panels provide descriptive seed-resolved validation and roster coverage for the Cu/Li/Na stress test. All ten archived DFT path records retain SCF-convergence warnings, so they support reproducibility and evidence-gating discussion but do not establish cross-element barrier accuracy or a second-material validation claim.

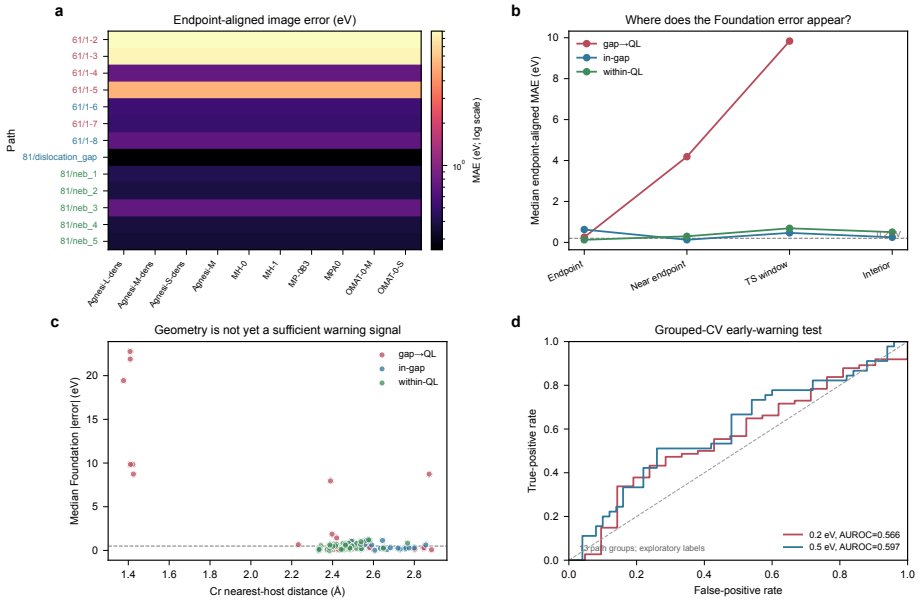


Fig. S24: **Reused unified diagnostic framework.** This exploratory Cr-61/Cr-81 page is retained as a provenance aid for separating system identity, path class, and evaluation operator. It mixes atom counts and path families and is not a pooled quantitative transfer result.

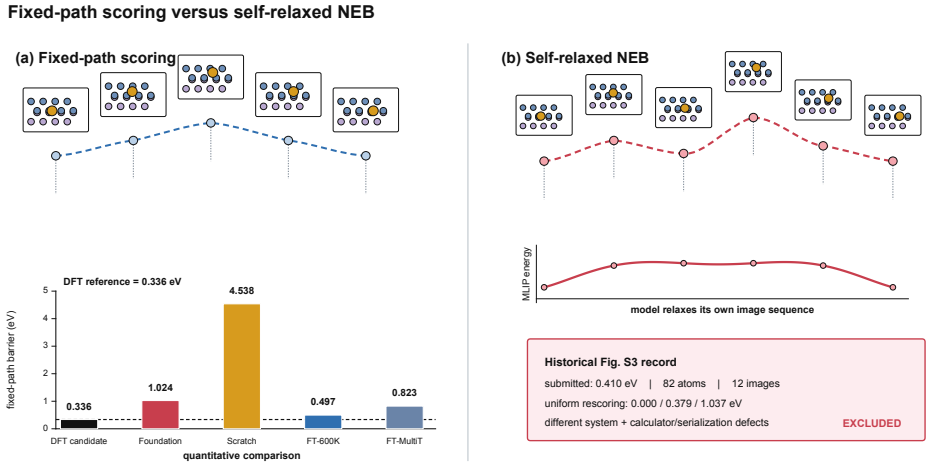


Fig. S25: Provenance audit of the historical self-consistent MLIP Nudged Elastic Band (NEB) record. (a) Fixed-geometry barrier assignments for the 61-atom, one-Cr within-gap probe used in the main-text comparison, with the current provisional QE candidate shown as the reference line. (b) Audit of the historical SI self-NEB record that reported 0.41 eV: it corresponds to a different 82-atom, two-Cr system with twelve images and contains calculator and final-band serialization defects. Uniform rescoring gives forward 0.00 eV, reverse 0.38 eV, and whole-band span 1.04 eV; the historical value is therefore excluded from quantitative ranking.

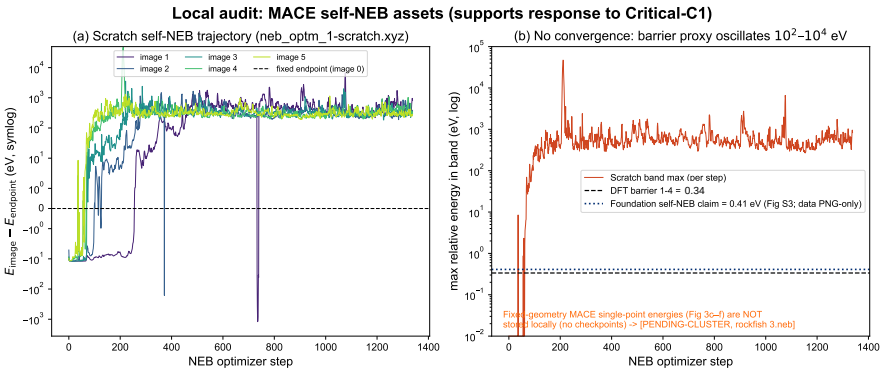


Fig. S26: **Local audit of self-consistent MLIP-NEB trajectory assets.** This diagnostic plot tracks the locally available scratch-model self-NEB trajectory and supports the separation between fixed-path DFT scoring and self-consistent MLIP-NEB stability. It is retained as an audit figure for Revision 1 traceability.

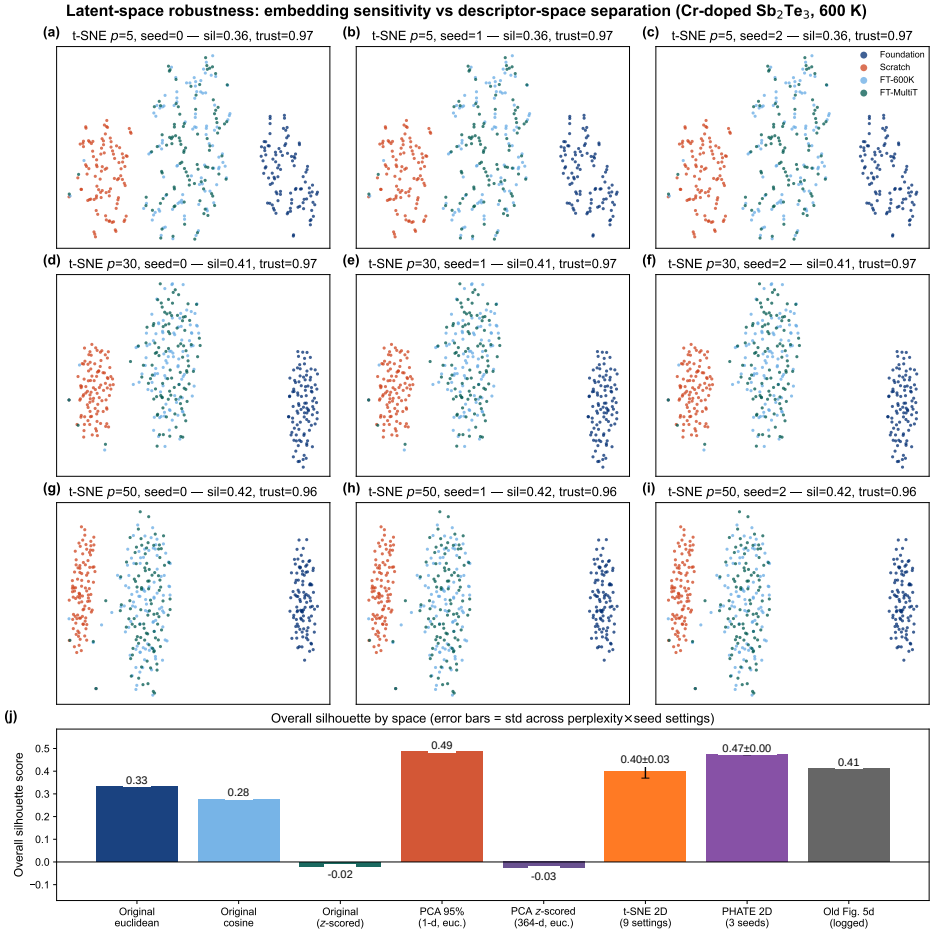


Fig. S27: **Latent-space robustness and original-descriptor checks.** The t-SNE panels show the audited perplexity range 0.36–0.42 and trustworthiness range 0.963–0.973; nominal seeds at fixed perplexity are identical under the locked PCA initialization. The PHATE summary is 0.47 ± 0.00. The summary panel compares these descriptive projection scores with original-descriptor-space silhouettes of 0.33 (raw Euclidean), 0.28 (cosine), and -0.02 (z-scored Euclidean).

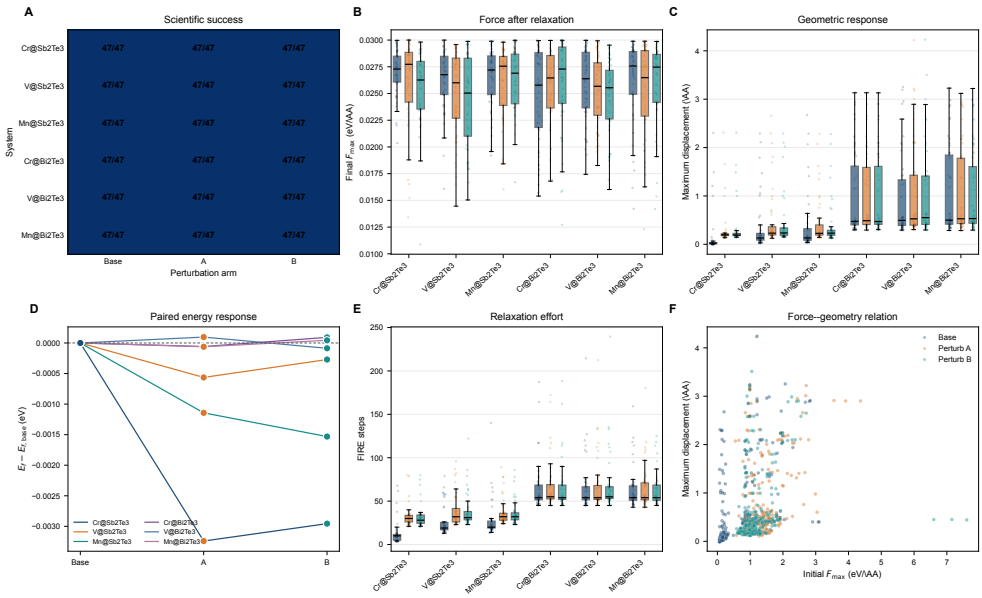


Fig. S28: **Campaign overview and relaxation outcomes.** (A) Scientific success across all 846 formal tasks. (B–C) Final-force and maximum-displacement distributions by perturbation arm. (D) Paired energy responses relative to the Base arm. (E) FIRE-step distributions. (F) Initial force versus final displacement. Points and distributions retain endpoint-level variation rather than showing only system medians.

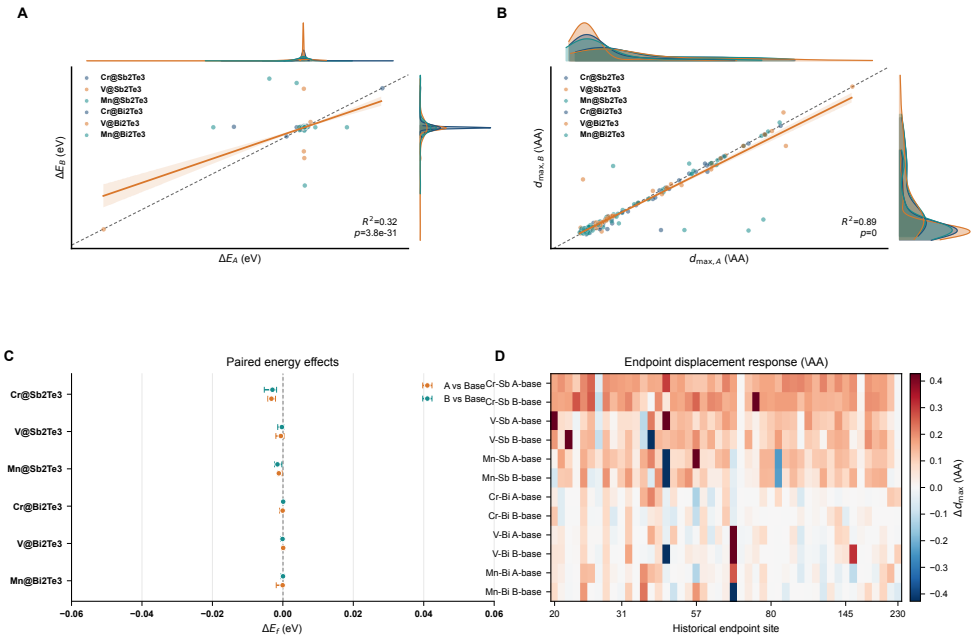


Fig. S29: **Perturbation response across systems and endpoints.** Joint distributions compare the energy and geometric response of Perturb A and B. Forest intervals summarize arm-specific changes relative to Base, and the endpoint heatmap exposes the site-level structure hidden by aggregate medians. Intervals are bootstrap summaries of the observed endpoint sample and should not be interpreted as uncertainty in the MACE potential itself.

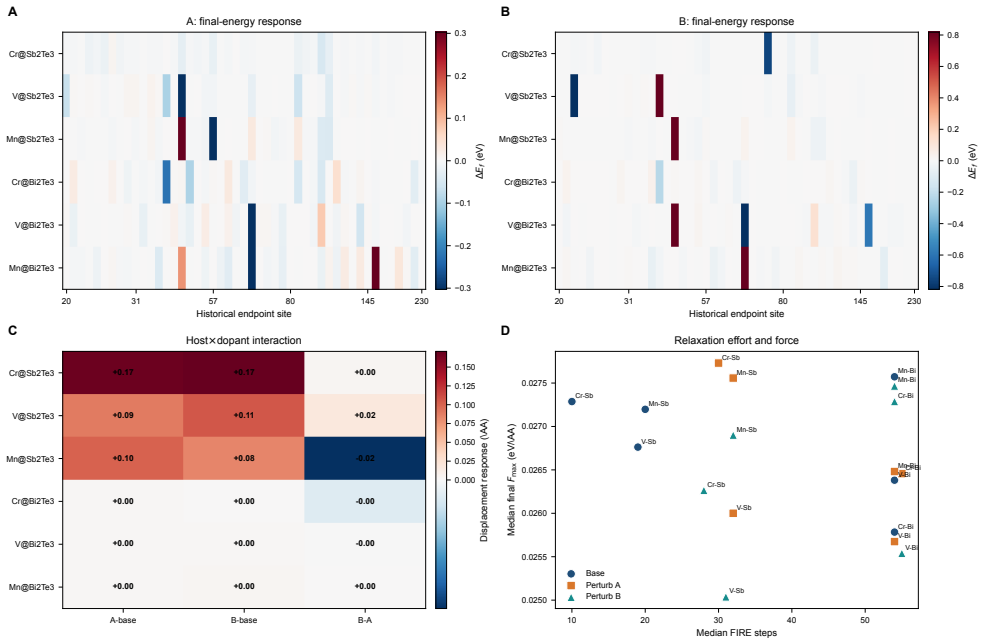


Fig. S30: **Host–dopant interactions and relaxation effort.** Endpoint heatmaps show energy response for each host–dopant system, while the interaction map compares the two perturbation arms. The effort panel relates initial force to the number of FIRE steps required by the fixed-cell relaxation protocol.